



THE THINGS
NETWORK



DATACAKE

First steps in IoT with LoRaWAN

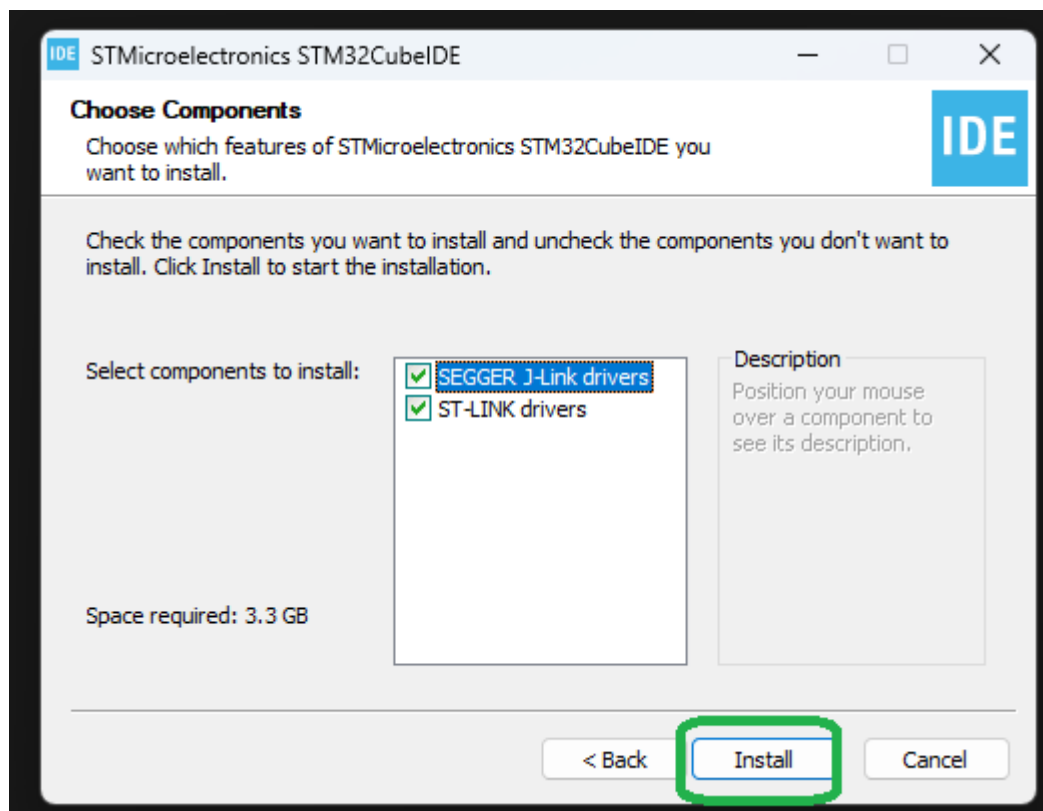
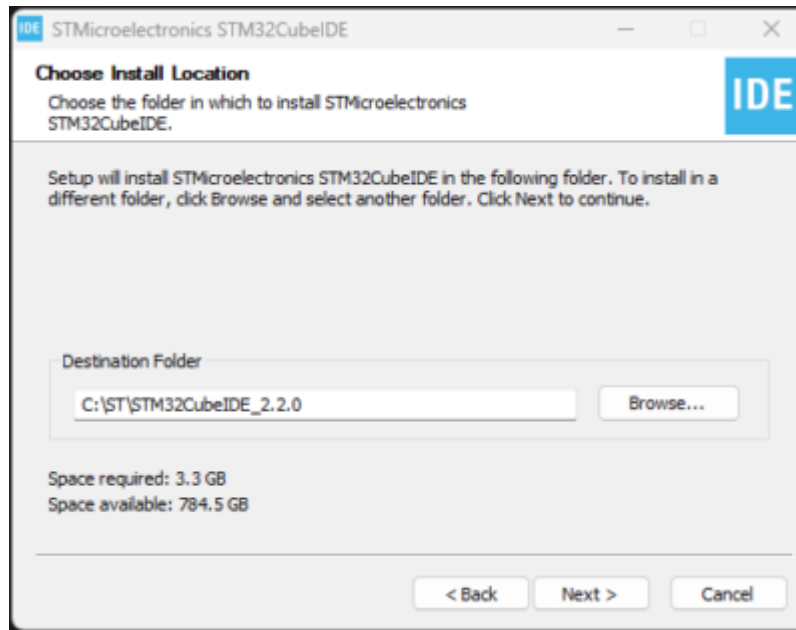
Development environment

- Hardware/Firmware
 - Gateway(optional): [NUCLEO-F746ZG](#) + [LRWAN GS HF1 LoRa](#)
 - Development board: [STM32WL55JC1](#) : STM32WL55JC1x Firmware: V1.3.1 + [driver](#)
 - Apiar IoT PCB: Available soon.
- *Software:*
 - [STMCubeIDE](#)(V:2.2.0)
 - [STM32CubeMX](#)(V:6.18.1)
 - [STMCubeWL](#)(V1.6.0/1: Library for STM32WL55JC
 - STM32WL55JC1x development board Firmware: V1.3.

Preparing the environment (toolchain)

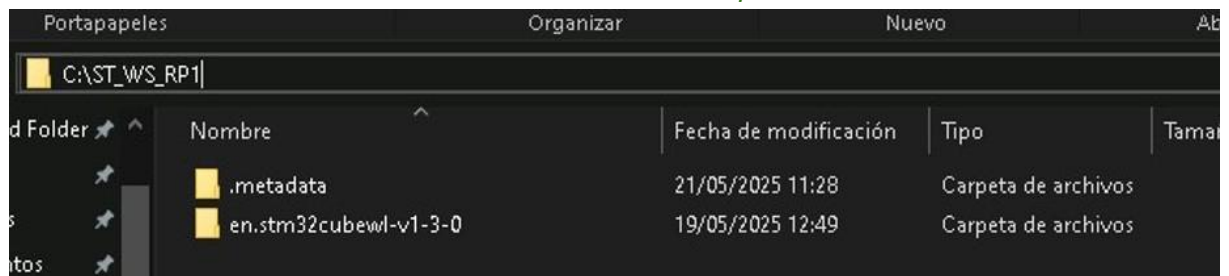
It is recommended to keep the same folder path and follow each step exactly.

- Download and install the integrated development environment (IDE) "STM32CubeIDE".

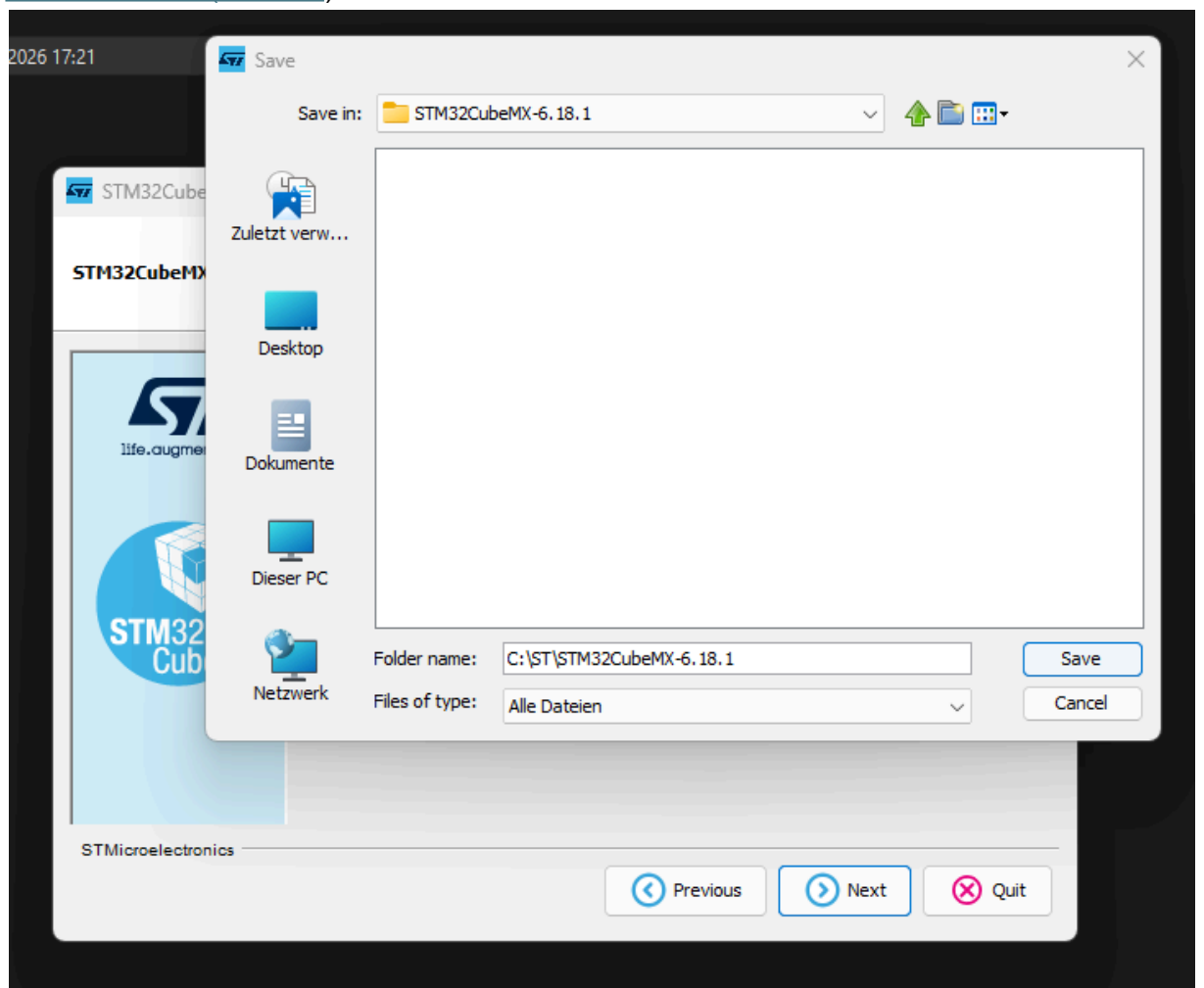


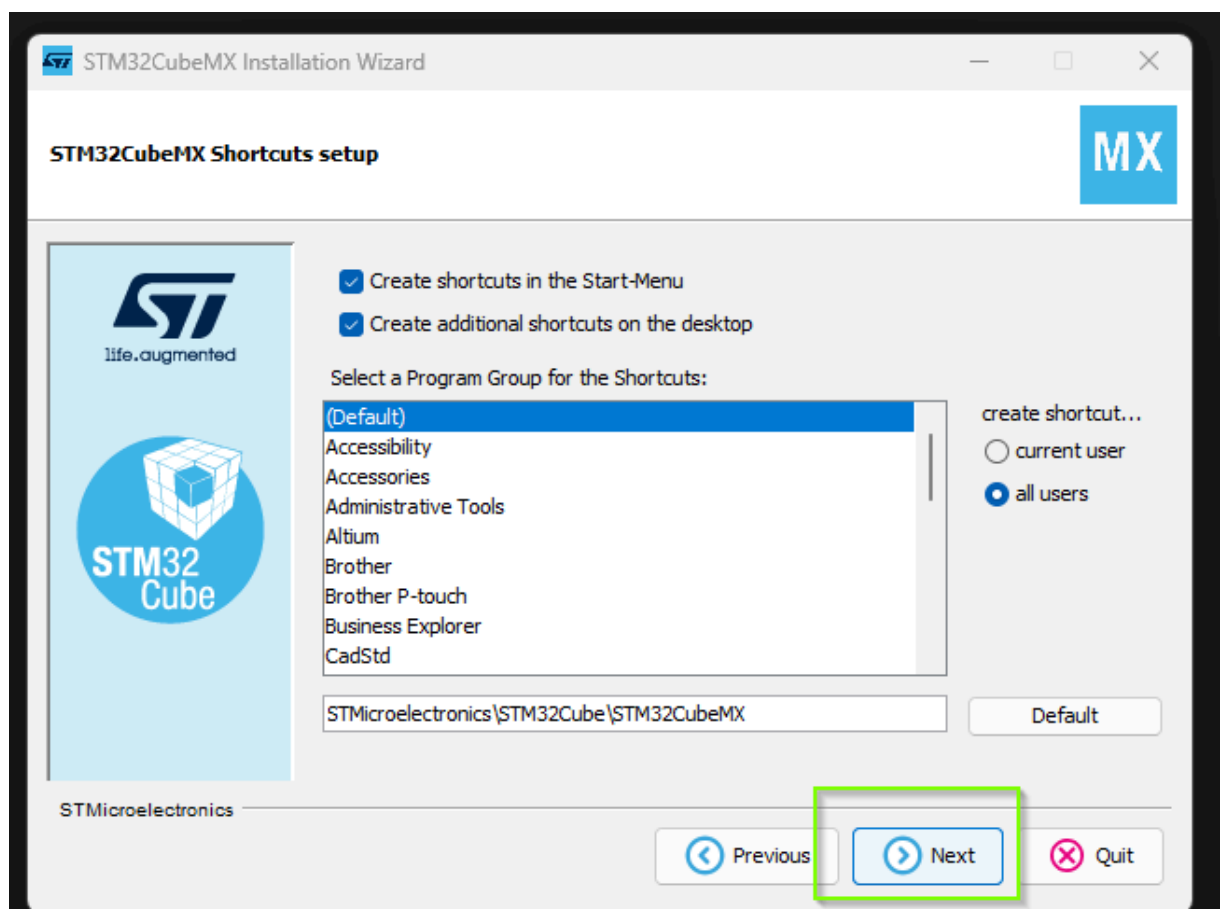
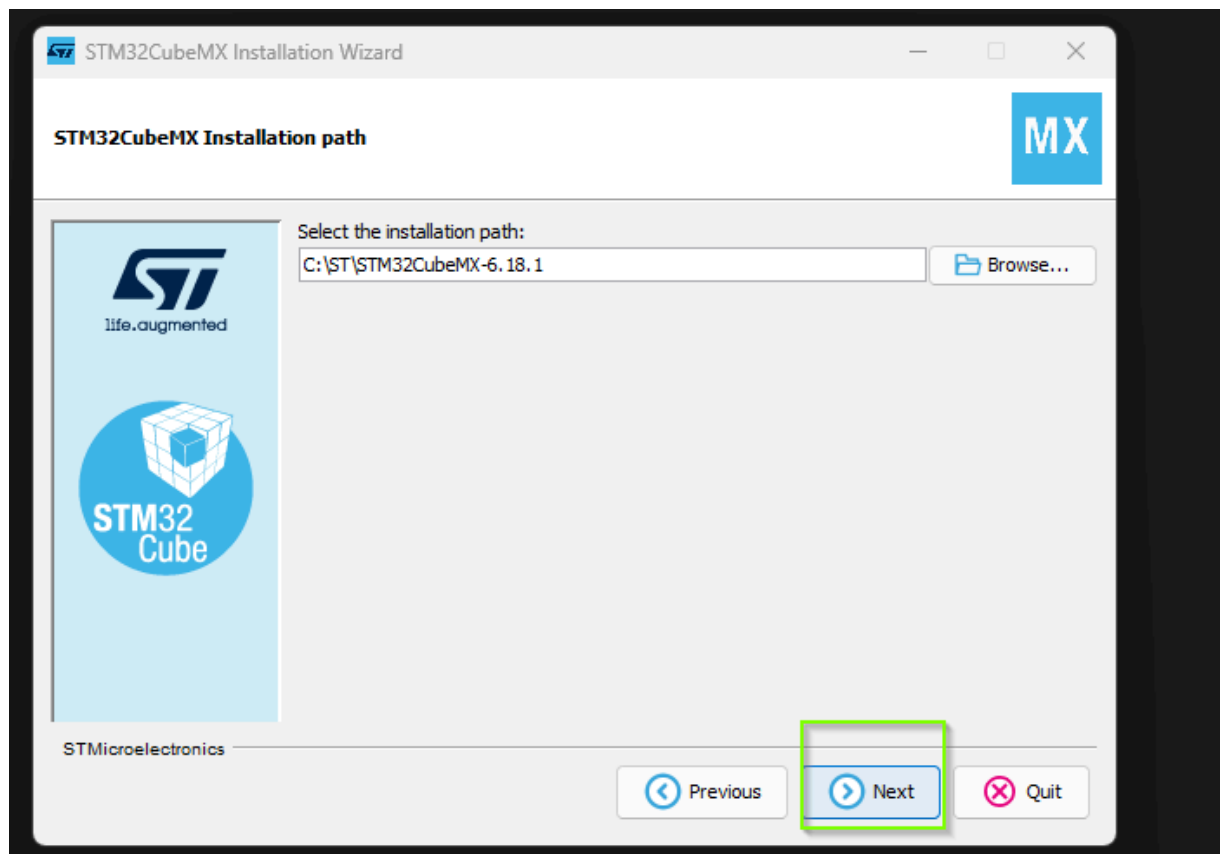
- Create the folder "C:\ST_WS_RP1".

It is recommended to use the same folder and the same path.

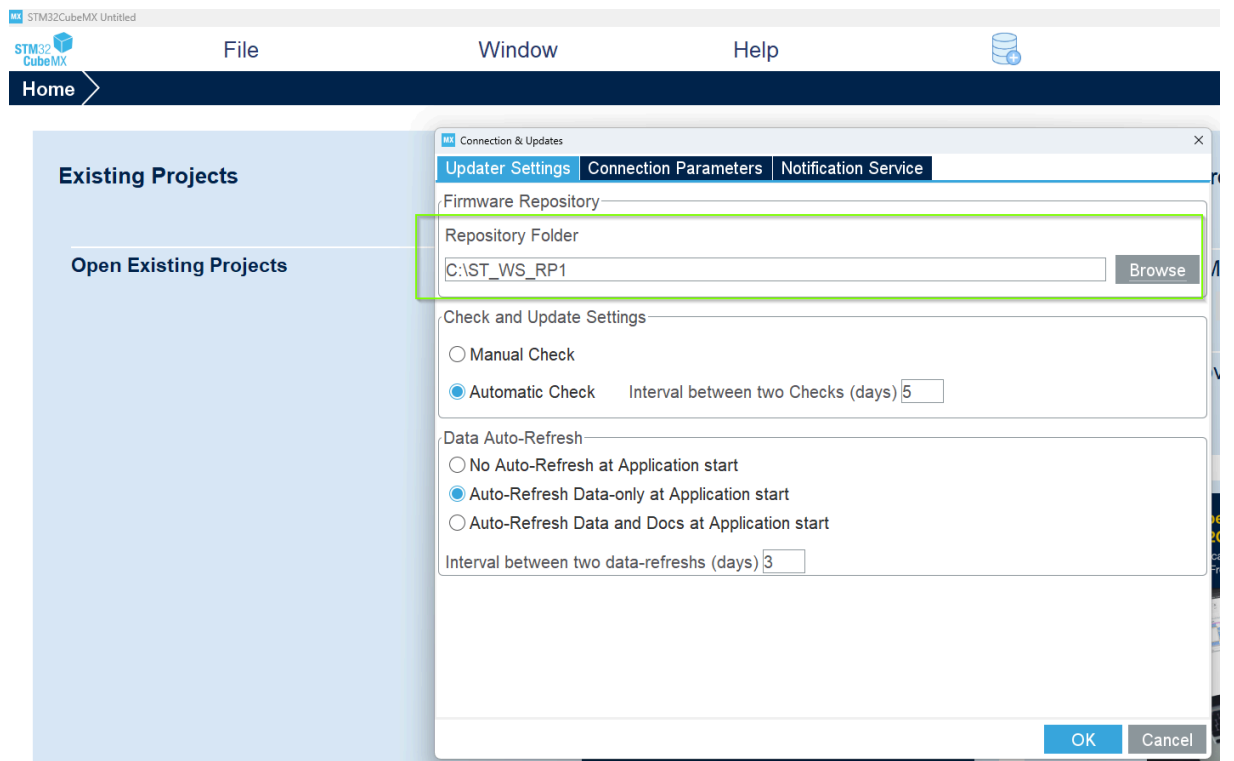
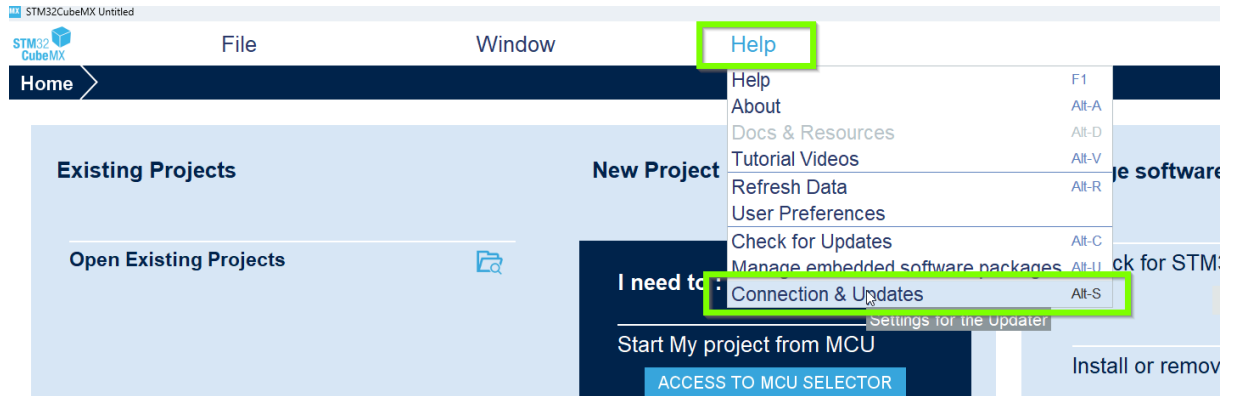


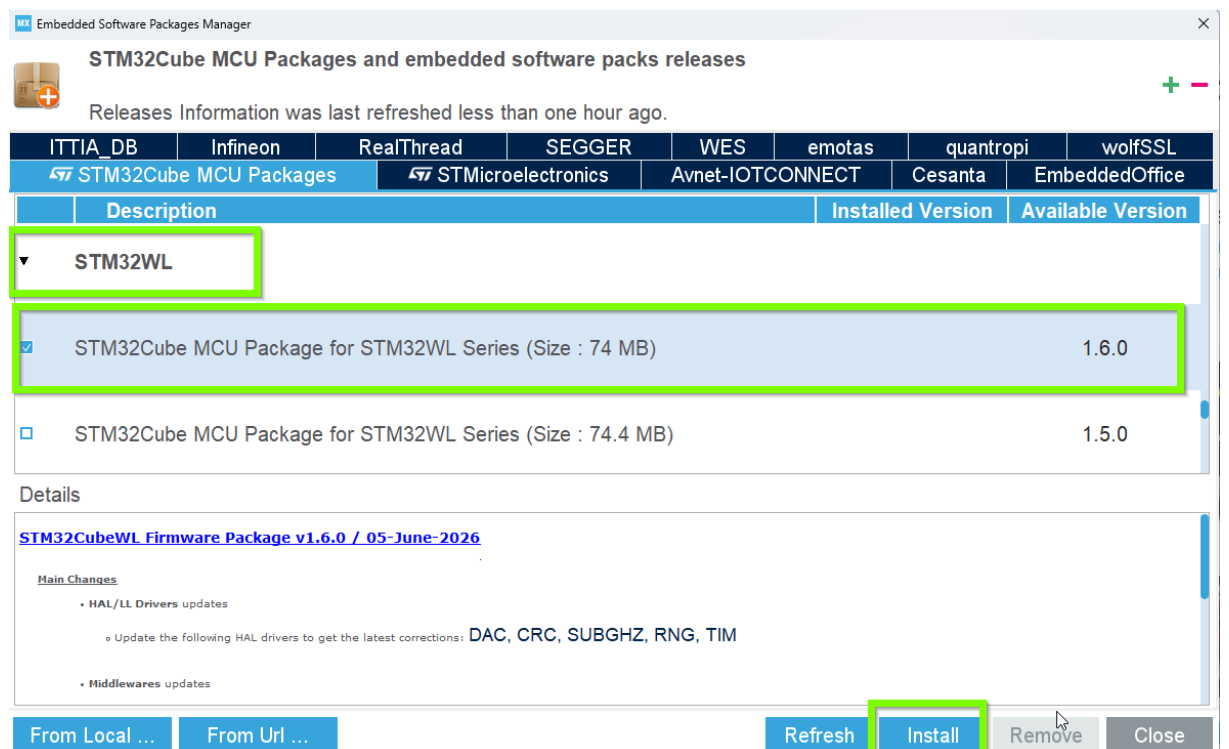
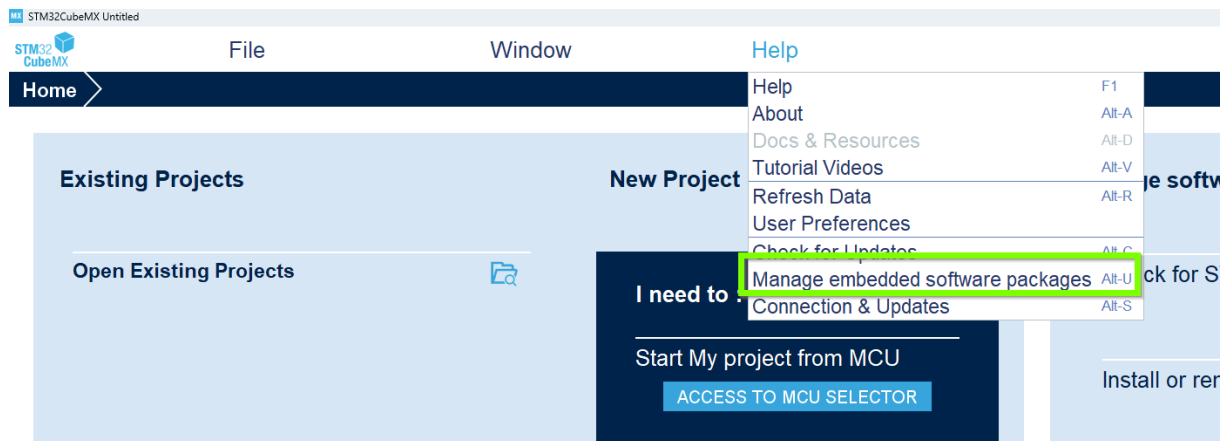
- Create the folder "C:\STSTM32CubeMX-6.18.1", Download and install STM32CubeMX(V:6.18.1).





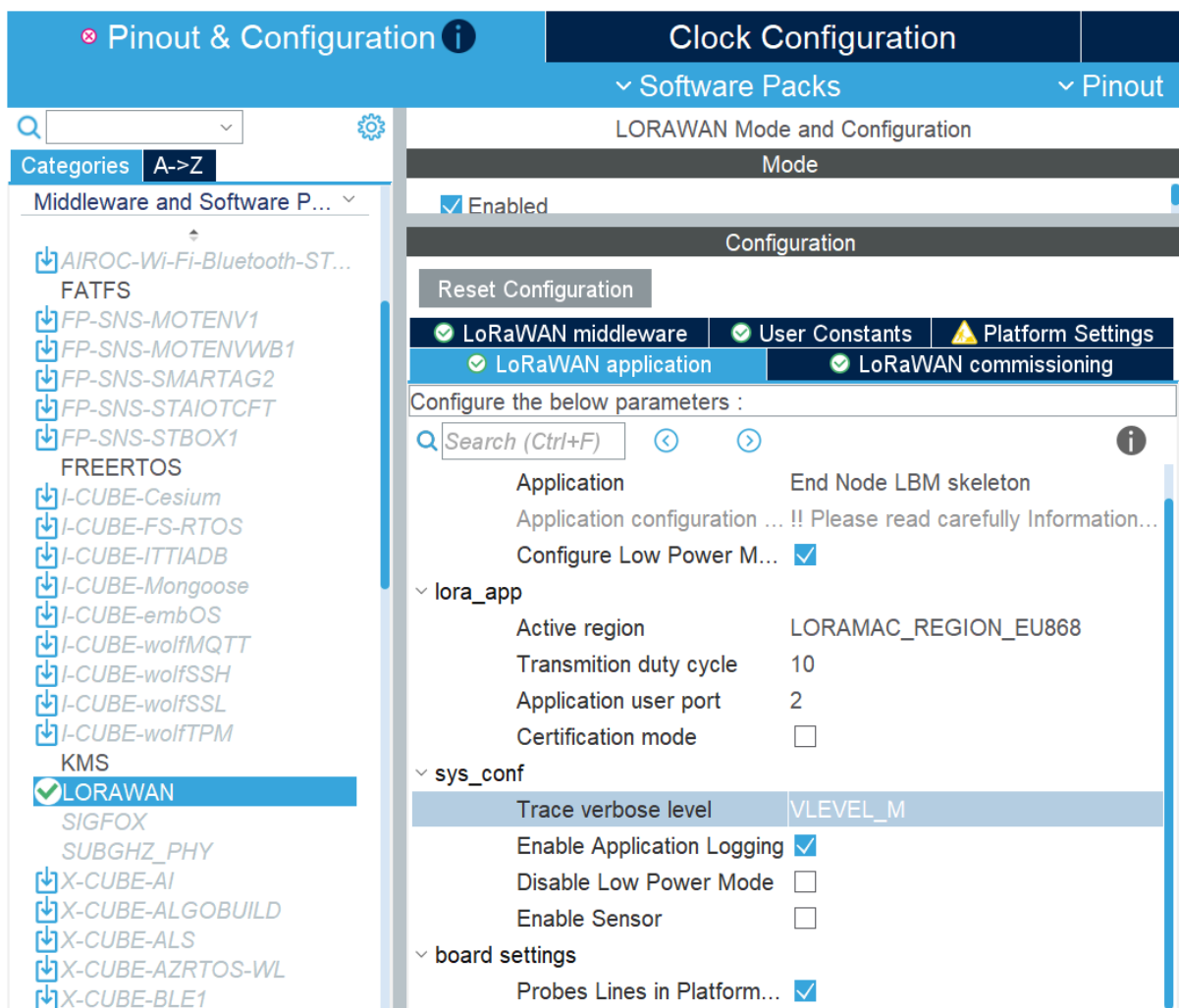
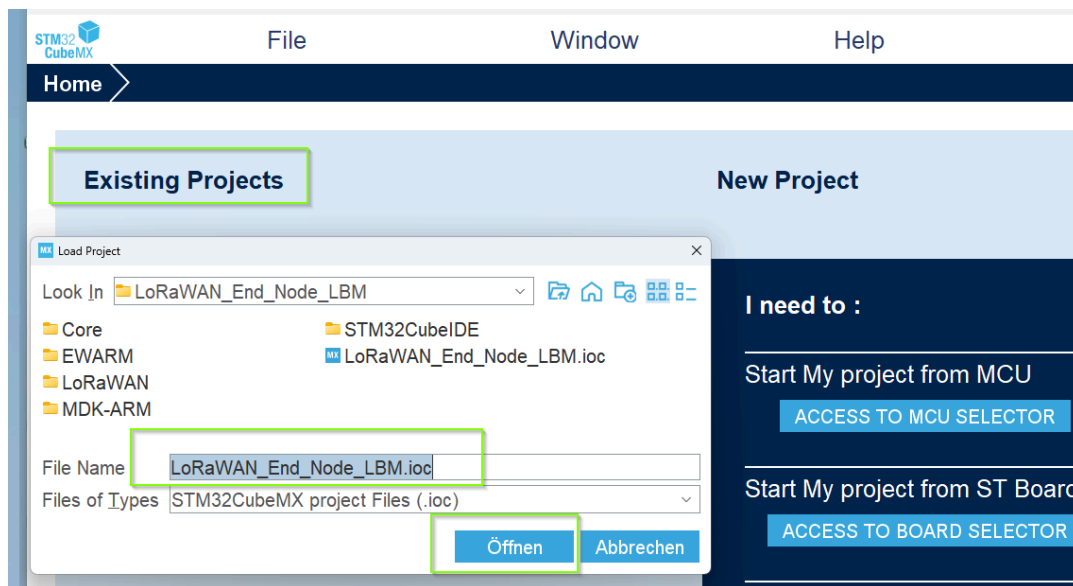
- Starting STM32CubeMX





LoRaWAN configuration:

- Starte:
LoRaWAN_End_Node_LBM.iocC:ST_WS_RP1STM32Cube_FW_WL_V1.6.0Projects
NUCLEO-WL55JCAApplicationsLoRaWANLoRaWAN_End_Node_LBMLoRaWAN_End
Node_LBM.ioc



LoRaWAN commissioning:

- “App/Join EUI”: Can be found on the board.

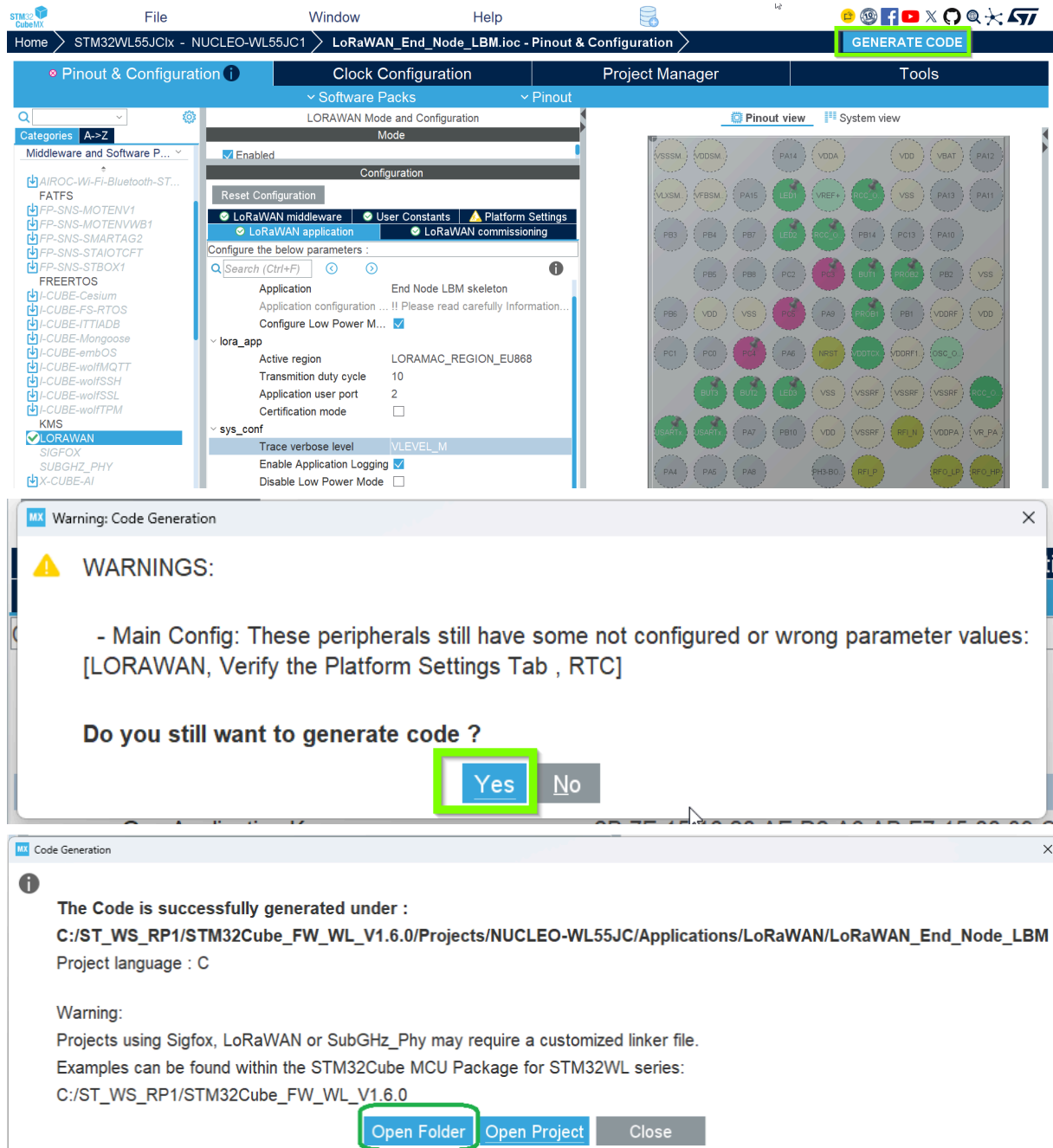


Important: The field "LORAWAN_device EUI" should NOT be defined as "00.00.00.00.00.00.00". Choose it yourself.

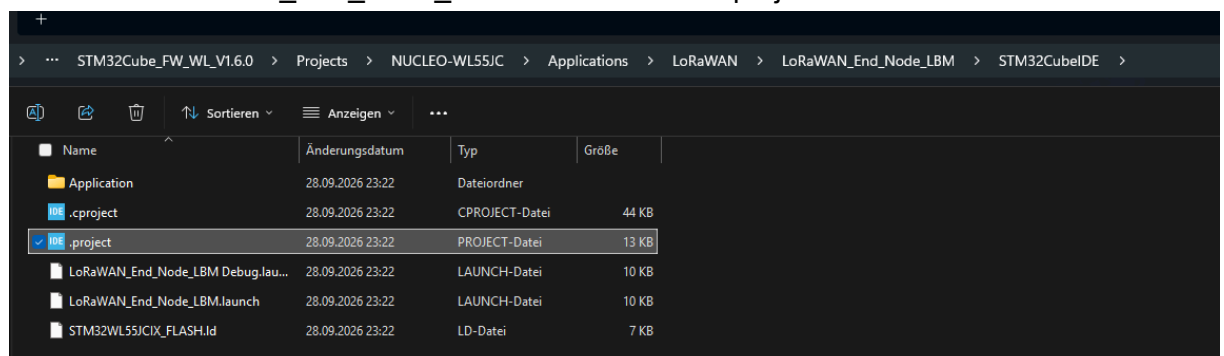
```
/*
 * End-device IEEE EUI (big endian)
 * When set to 00,00,00,00,00,00,00,00 DevEui is automatically set
 * with a value provided by MCU platform
 */
#define LORAWAN_DEVICE_EUI
00,00,00,00,00,00,00,00
```

- “Application key” and “Network key”: choose yourself

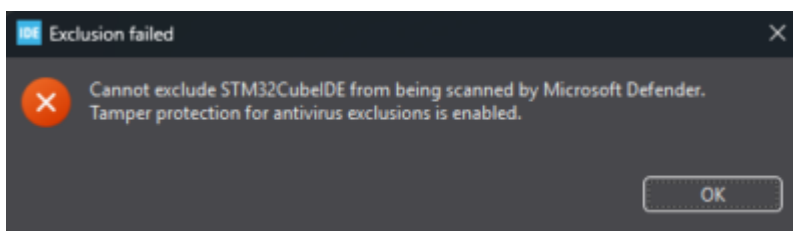
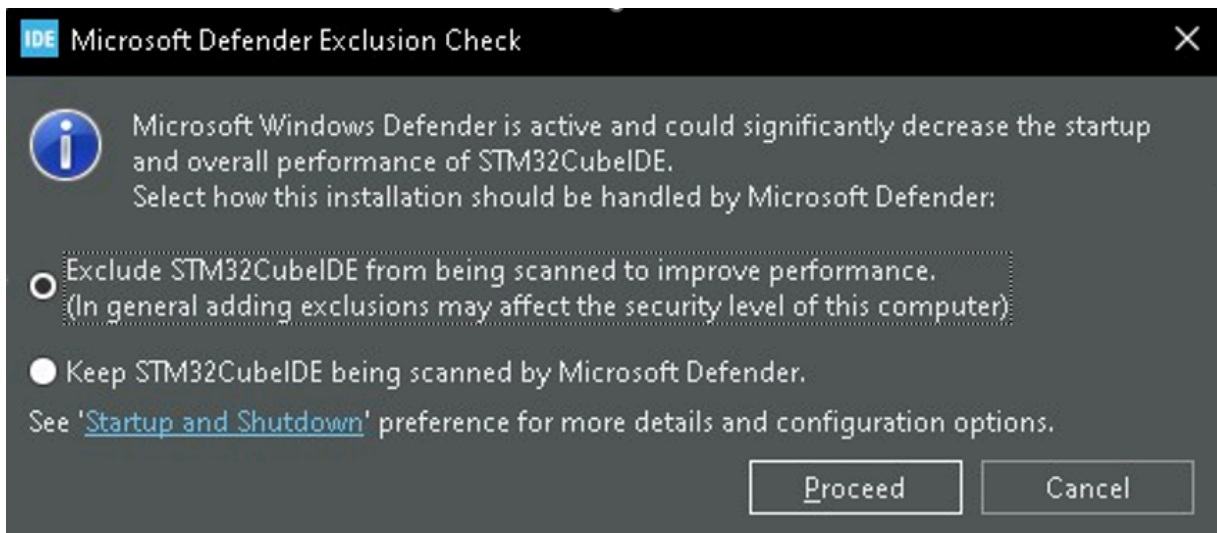
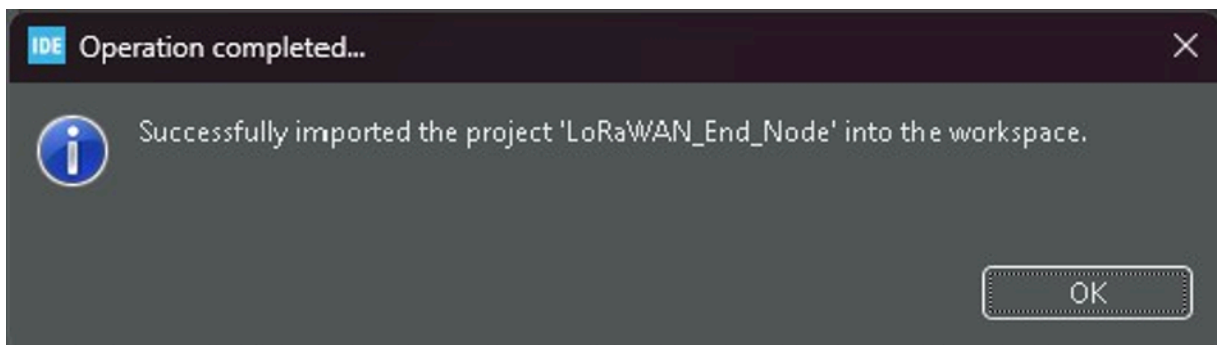
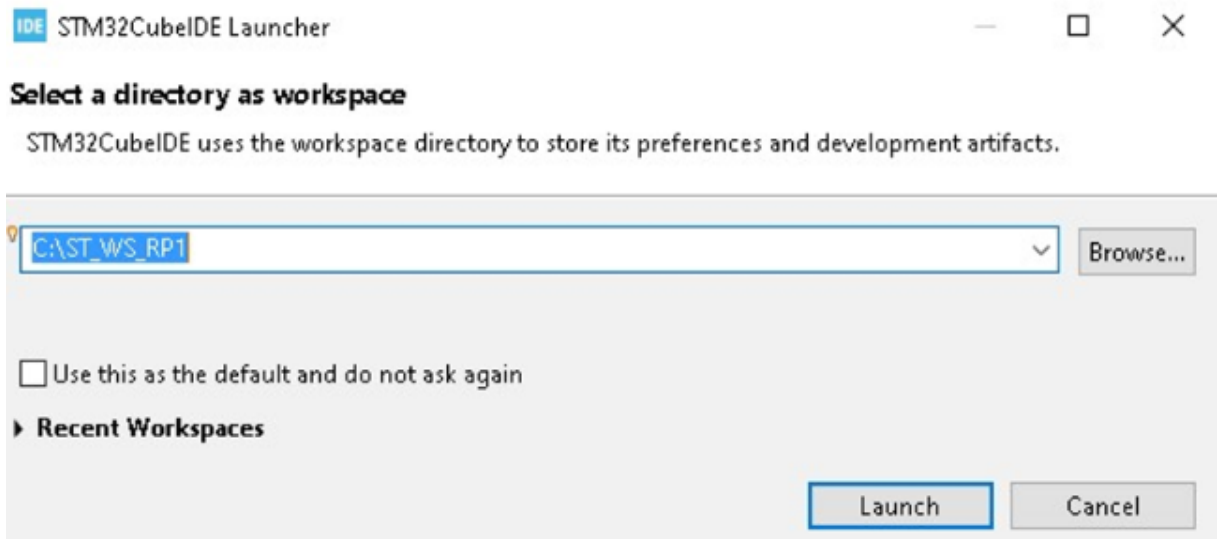
The screenshot shows the STM32CubeMX IDE interface for configuring a LoRaWAN project. The 'Pinout & Configuration' tab is active, displaying the 'LORAWAN Mode and Configuration' settings. The 'App/Join EUI' field is highlighted in green, showing the value '00,80,E1,15,05,00,D7,53'. The 'File' menu is open, and the 'Save Project' option is highlighted in green. The 'Save Project' option is also highlighted in the 'File' menu.



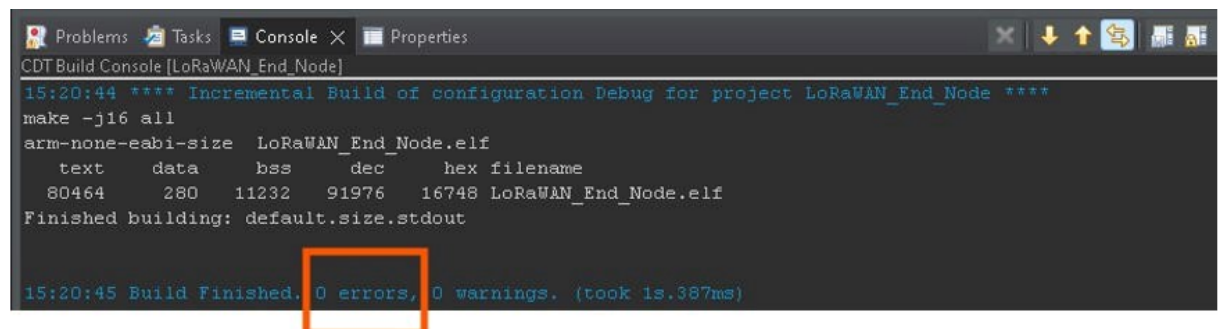
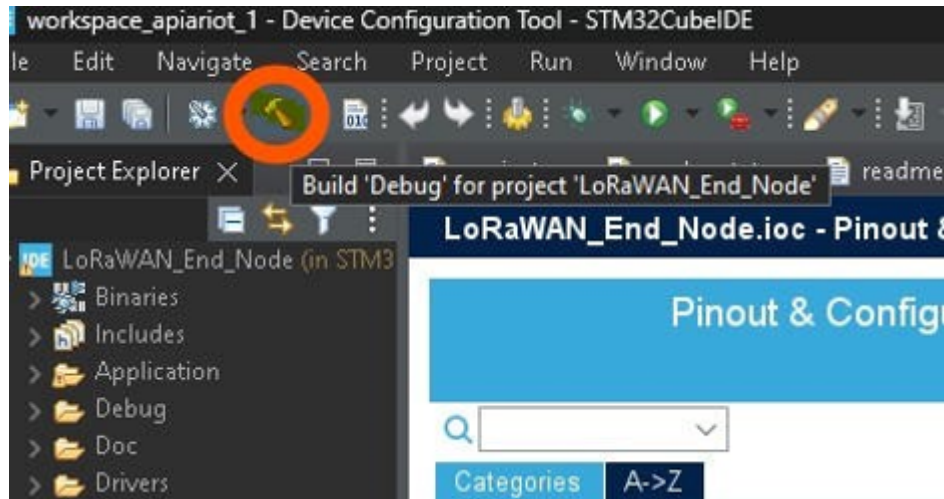
- Start the project with STM32cubeIDE
"C:ST_WS_RP1STM32Cube_FW_WL_V1.6.0ProjectsNUCLEO-WL55JCApplications
LoRaWANLoRaWAN_End_Node_LBMSTM32CubeIDE.project"



- Select C:\ST_WS_RP1 as the workspace.



- Compile the project without errors:



Programming code change

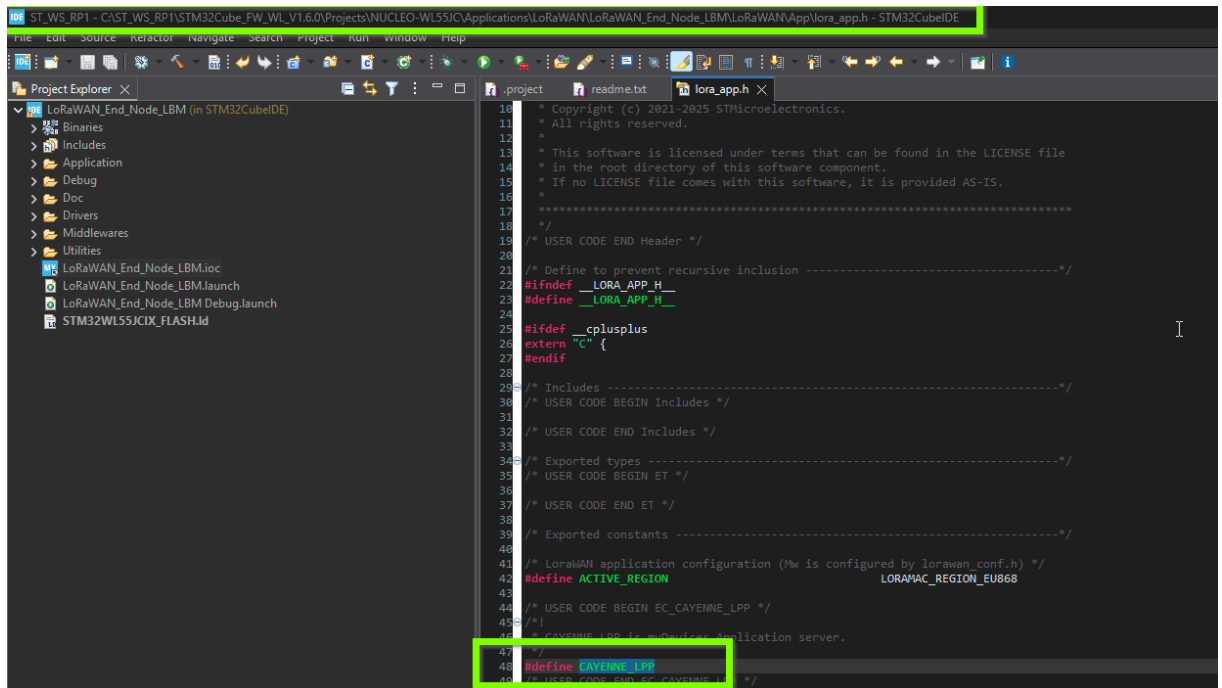
- Activate Cayenne's "Low Power Payload":
- Uncomment `/*#define CAYENNE_LPP` in `lora_app.h`*/:

#define CAYENNE_LPP

"C:\ST_WS_RP1\STM32Cube_FW_WL_V1.6.0\Projects\NUCLEO-WL55JC\Applications\LoRaWAN\LoRaWAN_End_Node_LBM\LoRaWANApp\lora_app.h"

A notice: Cayenne LPP is a compact data format for encoding sensor data for LoRaWAN. More information is available at this link:

[Was ist Cayenne Low Power Payload?](#) 🤔👤

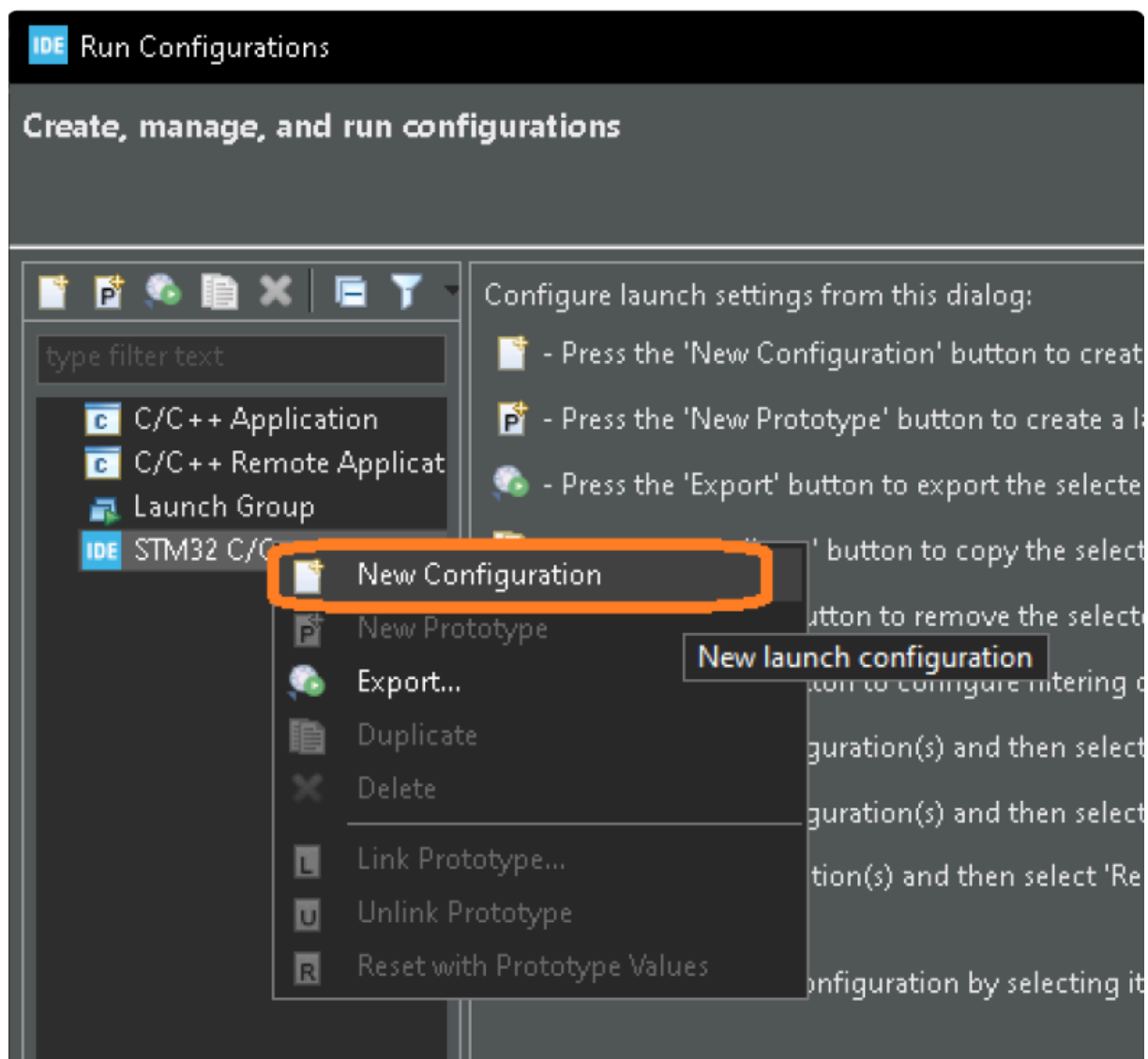
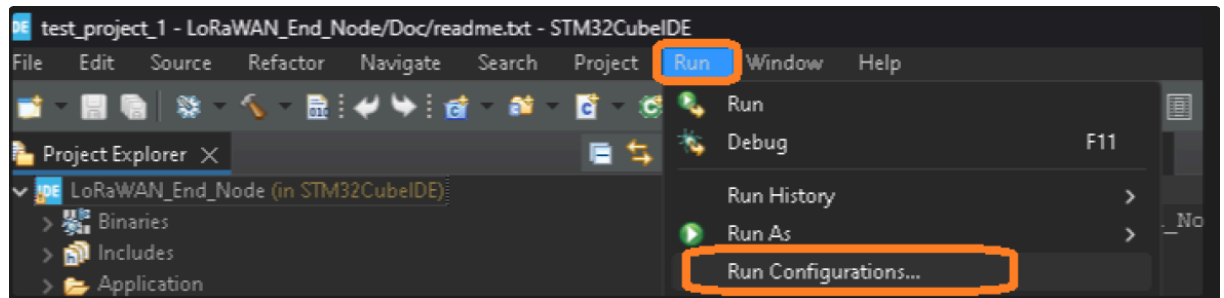


The screenshot shows an IDE window with the file `lora_app.h` open. The Project Explorer on the left shows the project structure. The main editor displays the code for `lora_app.h`. The line `#define CAYENNE_LPP` is highlighted with a green box, indicating it has been uncommented. The code includes various headers and defines, such as `LORAMAC_REGION_EU868` and `LORAMAC_REGION_US915`.

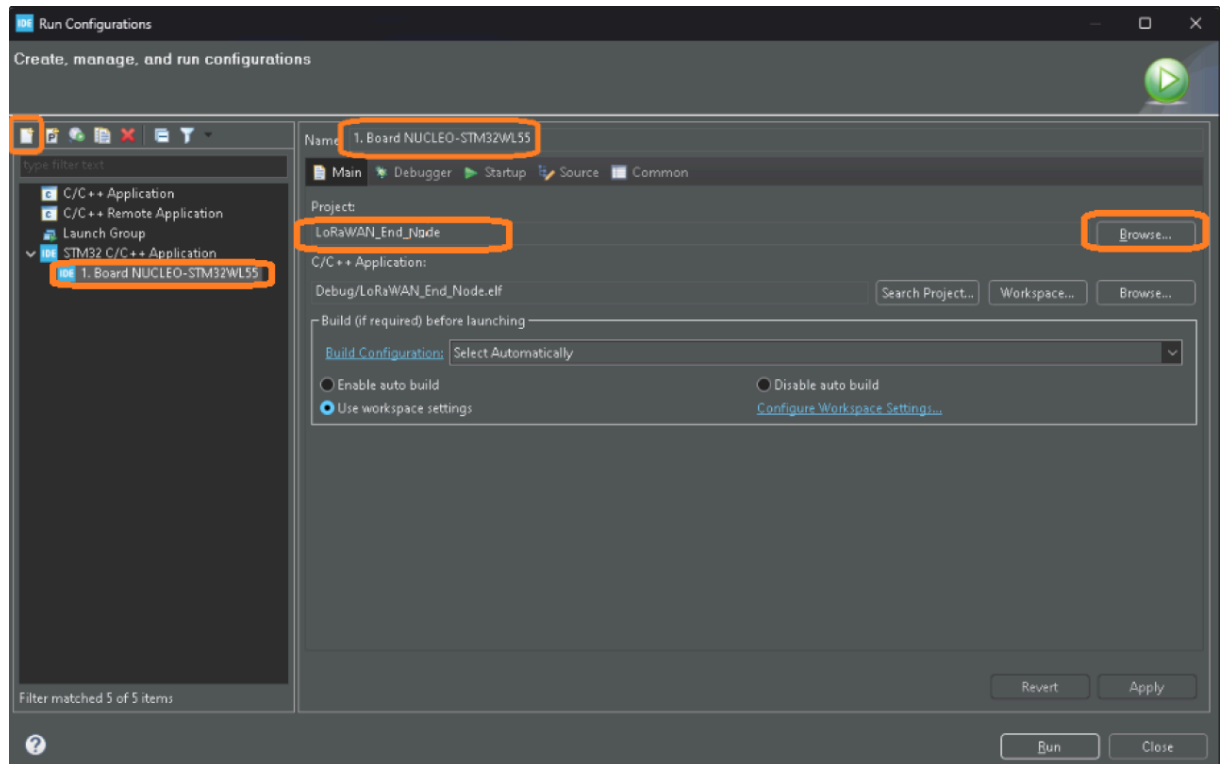
```
10  * Copyright (c) 2021-2025 STMicroelectronics.
11  * All rights reserved.
12  *
13  * This software is licensed under terms that can be found in the LICENSE file
14  * in the root directory of this software component.
15  * If no LICENSE file comes with this software, it is provided AS-IS.
16  *
17  * =====
18  */
19  /* USER CODE END Header */
20
21  /* Define to prevent recursive inclusion -----*/
22  #ifndef LORA_APP_H
23  #define LORA_APP_H
24
25  #ifdef __cplusplus
26  extern "C" {
27  #endif
28
29  /* Includes -----*/
30  /* USER CODE BEGIN Includes */
31
32  /* USER CODE END Includes */
33
34  /* Exported types -----*/
35  /* USER CODE BEGIN ET */
36
37  /* USER CODE END ET */
38
39  /* Exported constants -----*/
40
41  /* LoRaWAN application configuration (Mw is configured by lorawan_conf.h) */
42  #define ACTIVE_REGION LORAMAC_REGION_EU868
43
44  /* USER CODE BEGIN EC_CAYENNE_LPP */
45  /*
46  * CAYENNE_LPP is a compact data format for encoding sensor data.
47  */
48  #define CAYENNE_LPP
49  /* USER CODE END EC_CAYENNE_LPP */
```

Board settings

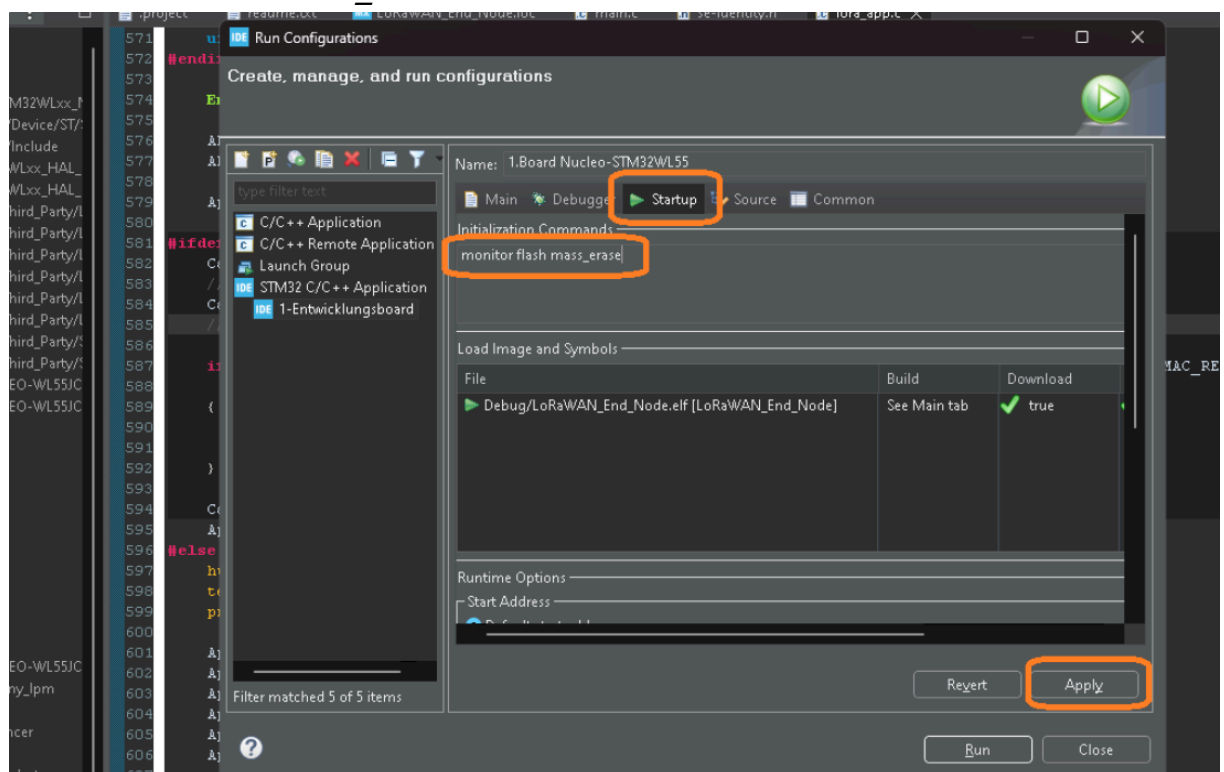
- Create a new configuration for the development board.



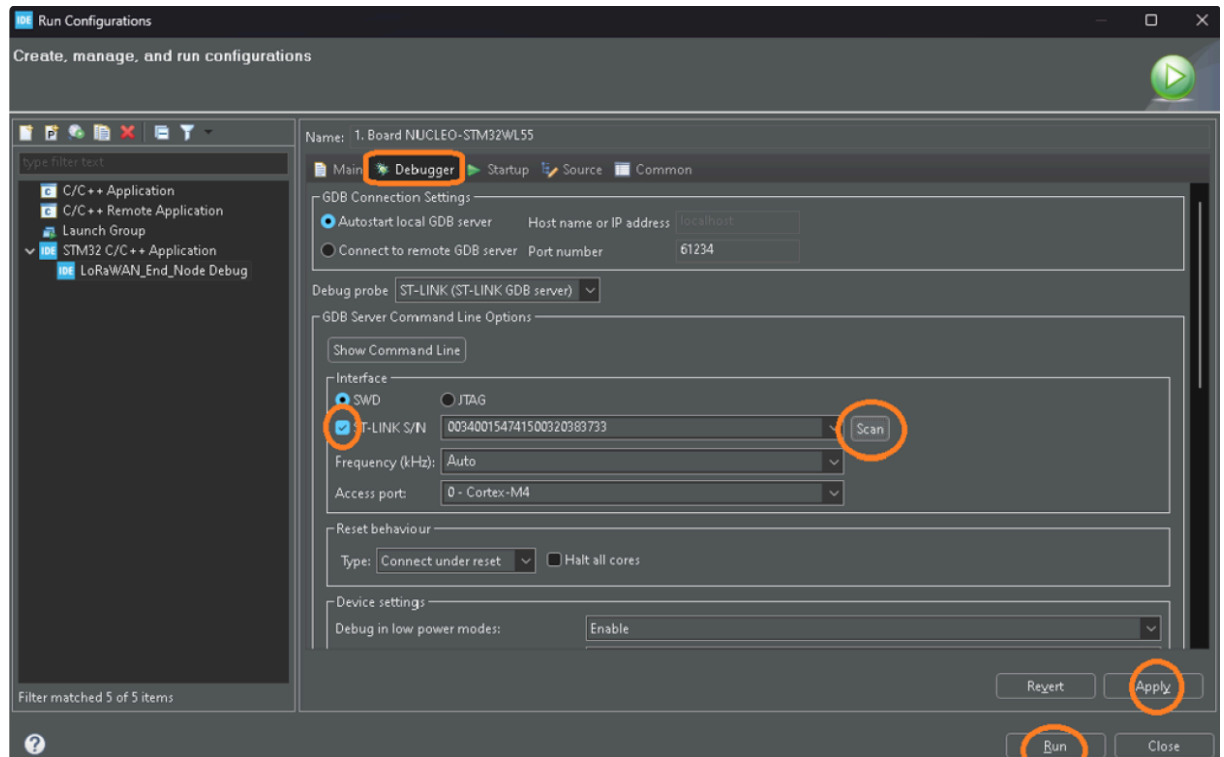
- Name your board NUCLEO-STIM32WL55



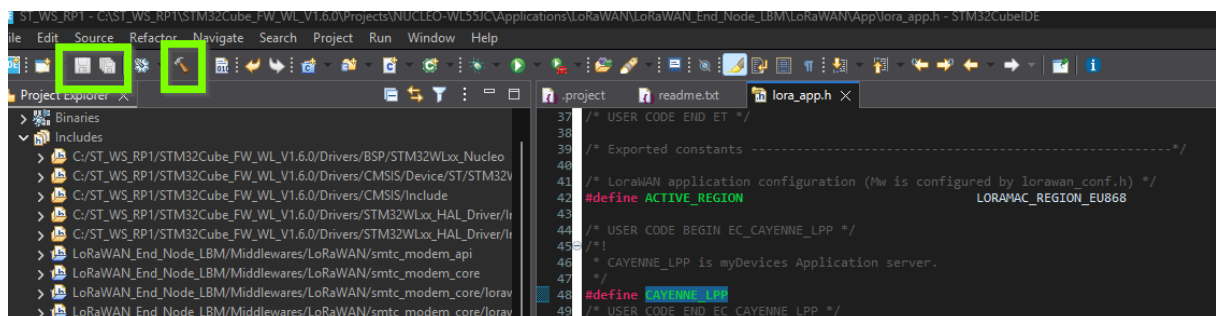
- Add under “*Initialization Commands*” Add the following command:
“*monitor flash mass_erase*”



- Connect the development board to the PC via USB

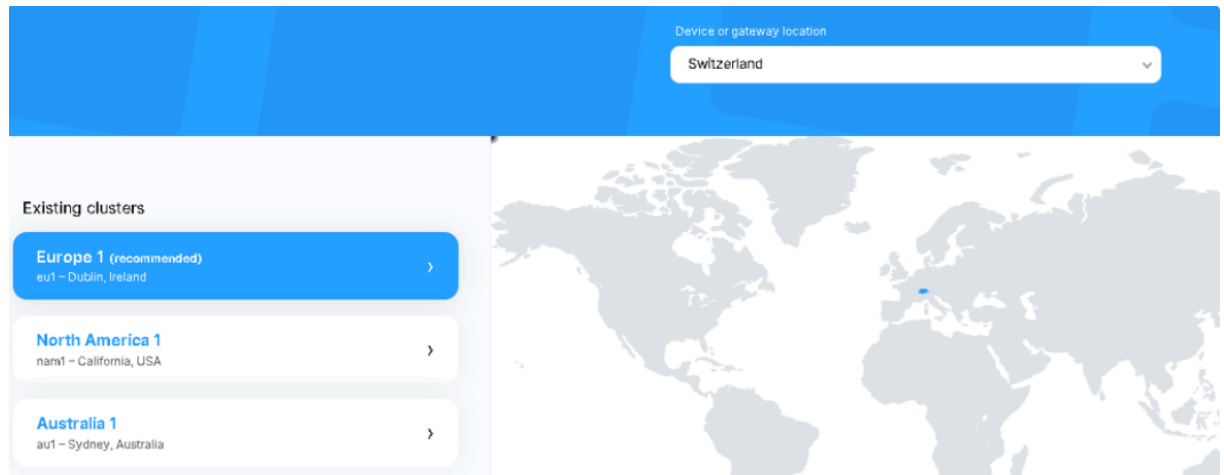


- Save the configuration



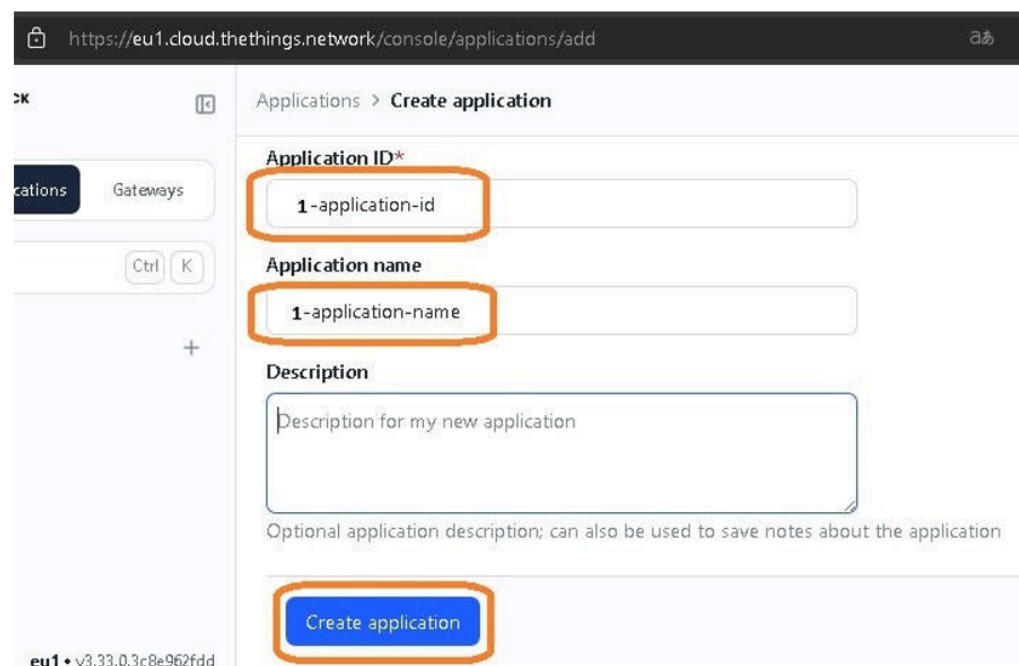
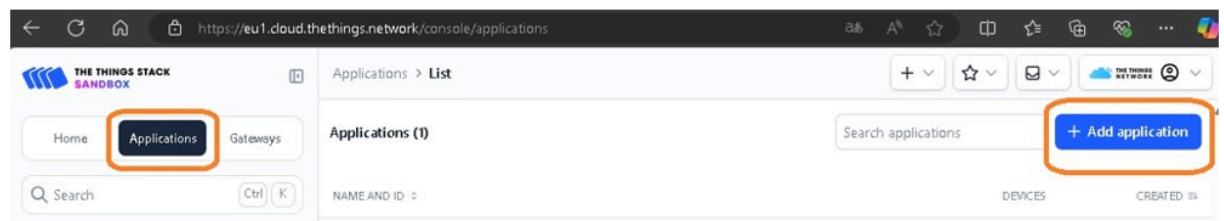
The Thing Network (TTN) / Stacks

- [Create](#) an account to access [The things network](#) to obtain.
- Select Cluster Europe 1 (For Switzerland)



Cluster Für die Schweiz: Europe 1

- Create an application



Create application

Within applications, you can register and manage end devices and their network data. After setting up your device fleet, use one of our many integration options to pass relevant data to your external services.

Learn more in our guide on [Adding Applications](#).

Application ID*

my-application-1-test

Application name

my application 1 test

Description

Description for my new application

Optional application description; can also be used to save notes about the application

Application properties

Labels

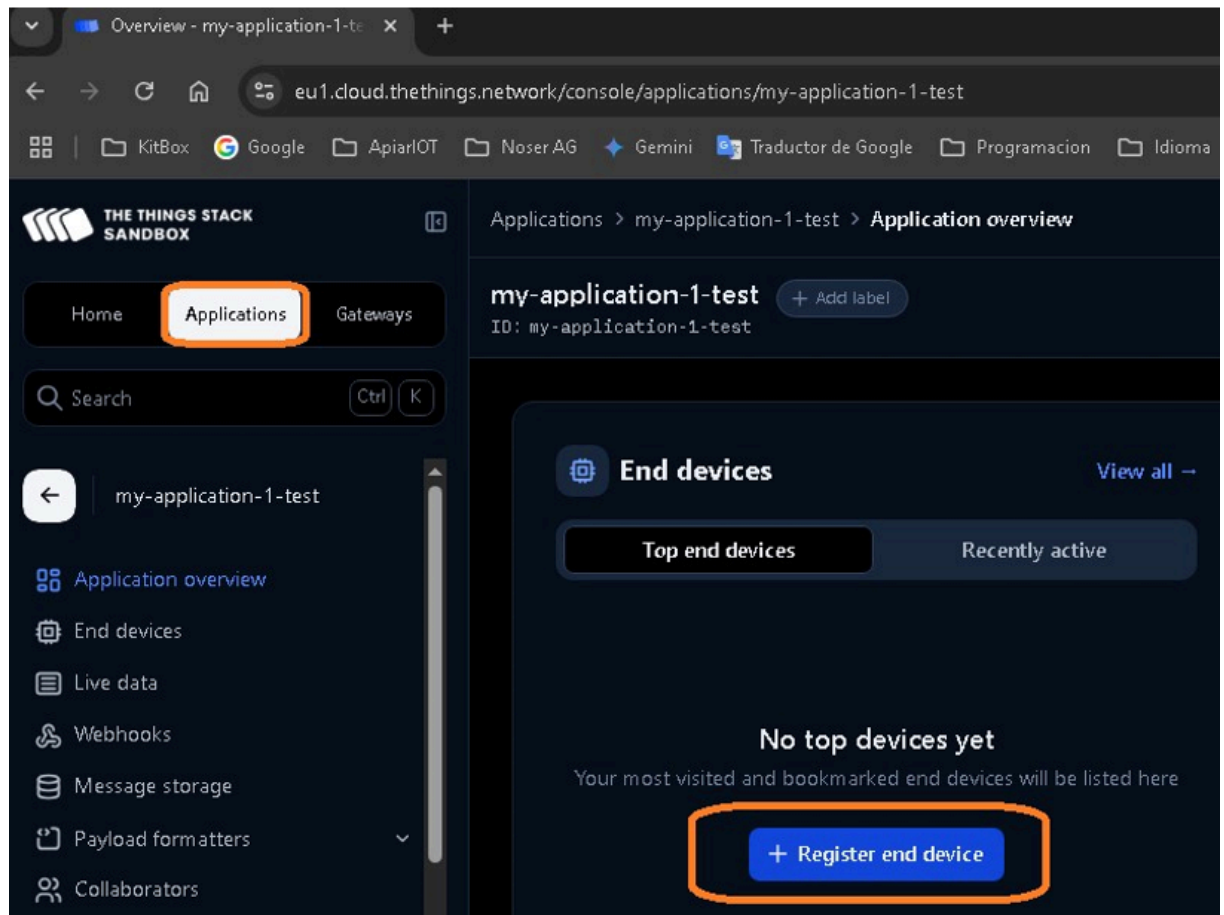
Select...



Add a label or [create one](#) to categorize your applications

Create application

- Register the "end device".



- End device type, End device brand, Model, Hardware version, Firmware version, Profile region, Frequency plan

Register end device

Does your end device have a LoRaWAN® Device Identification QR Code? Scan it to speed up onboarding.

 Scan end device QR code

 [Device registration help](#)

End device type

Input method

☒ Select the end device in the LoRaWAN Device Repository

☐ Enter end device specifics manually

End device brand  *

STMicroelectronic...

Model  *

NUCLEO-WL55J...

Hardware Ver.  *

1.0

Firmware Ver.  *

LoRaWA...

Profile (Region) *

EU_863_870

NUCLEO-WL55JC1

LoRaWAN Specification 1.0.3, RP001 Regional Parameter (OTAA), Class A



The STMicroelectronics NUCLEO-WL55JC1 is a development board based on the STM32WL55JC microcontroller that comes with a built-in temperature sensor. The NUCLEO-WL55JC1 board supports a wide choice of Integrated Development Environments (IDE) and an expansion connector for Arduino UNO V3. Suitable for rapid prototyping by adding sensors and actuators.

[Product website](#) | [Data sheet](#)

LoRaWAN_End_v1.0.0

LoRaWAN_End_N v1.1.0

LoRaWAN_End_N v1.2.0

Other...

Frequency plan  *

Europe 863-870 MHz (SF9 for RX2 - recommended)

- Search the information on "LoRaWAN commissioning"

Applications > my-application-1-test > End devices > mdfsdfdf > Device overview

mdfsdfdsf ID: mdfsdfdf + Add label No activity yet • ↑↓ n/a up / n/a down

Device overview Live data Messaging Location Payload formatters **Settings**

End device info

Device repository →



NUCLEO-WL55JC1
STMicroelectronics International NV

Device website Data sheet

Light Pressure Temperature Humidity

Latest decoded payload

See in live data →

No recent activity
There haven't been any uplinks recently in this application

device-3-d87e ID: device-3-d87e + Add label No activity yet • ↑↓ n/a up / n/a down

Network layer

LoRaWAN network-layer settings, behavior and session

Frequency plan ⓘ *

Europe 863-870 MHz (SF9 for RX2 - recommended) | ▾

LoRaWAN version ⓘ *

LoRaWAN Specification 1.0.1 | ▾

Regional Parameters version ⓘ *

TS001 Technical Specification 1.0.1 | ▾

LoRaWAN class capabilities ⓘ

☐ Supports class B

☐ Supports class C

Activation mode ⓘ *

☒ Over the air activation (OTAA)

☐ Activation by personalization (ABP)

☐ Define multicast group (ABP & Multicast)

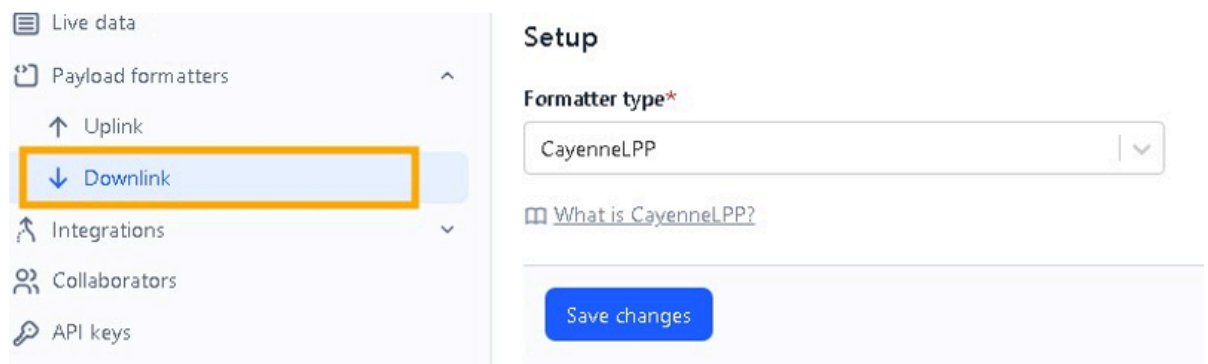
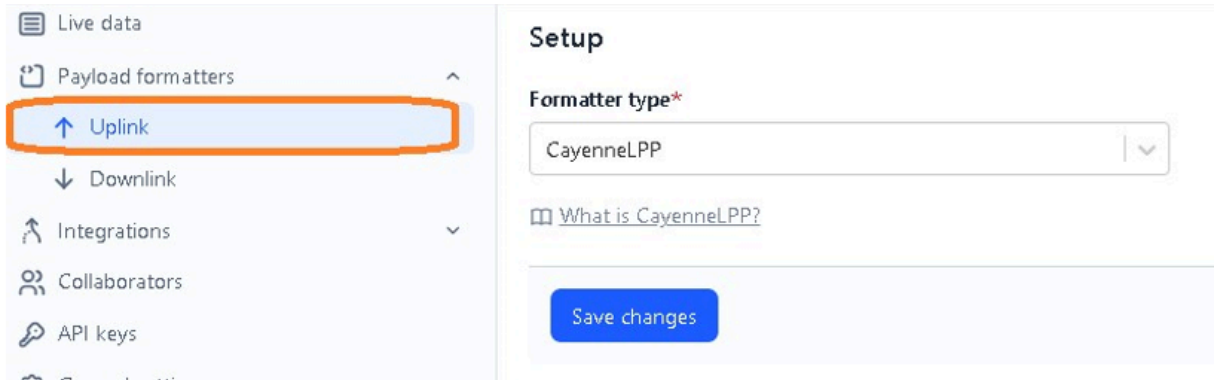
Session and MAC state reset ⓘ

Reset session and MAC state

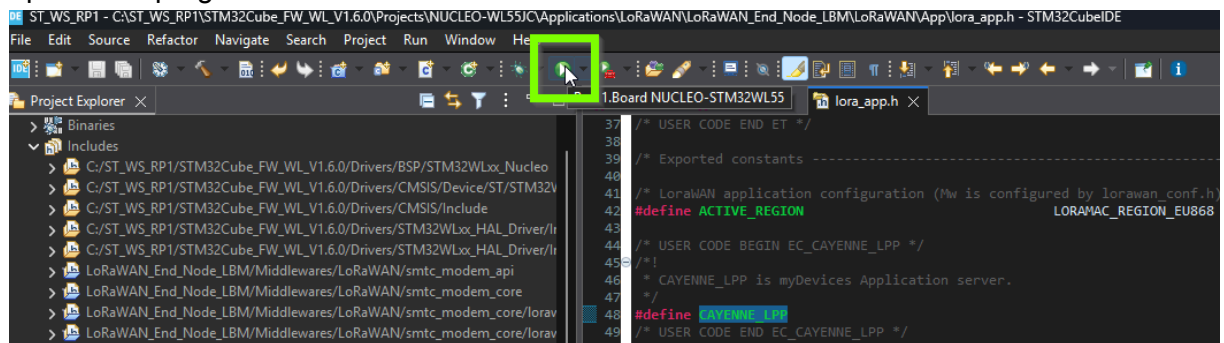
Advanced MAC settings ▾

Save changes

- Configure the "Uplink" and the "Downlink" for the "Payload" of the "End-device".



- Upload the program to the board

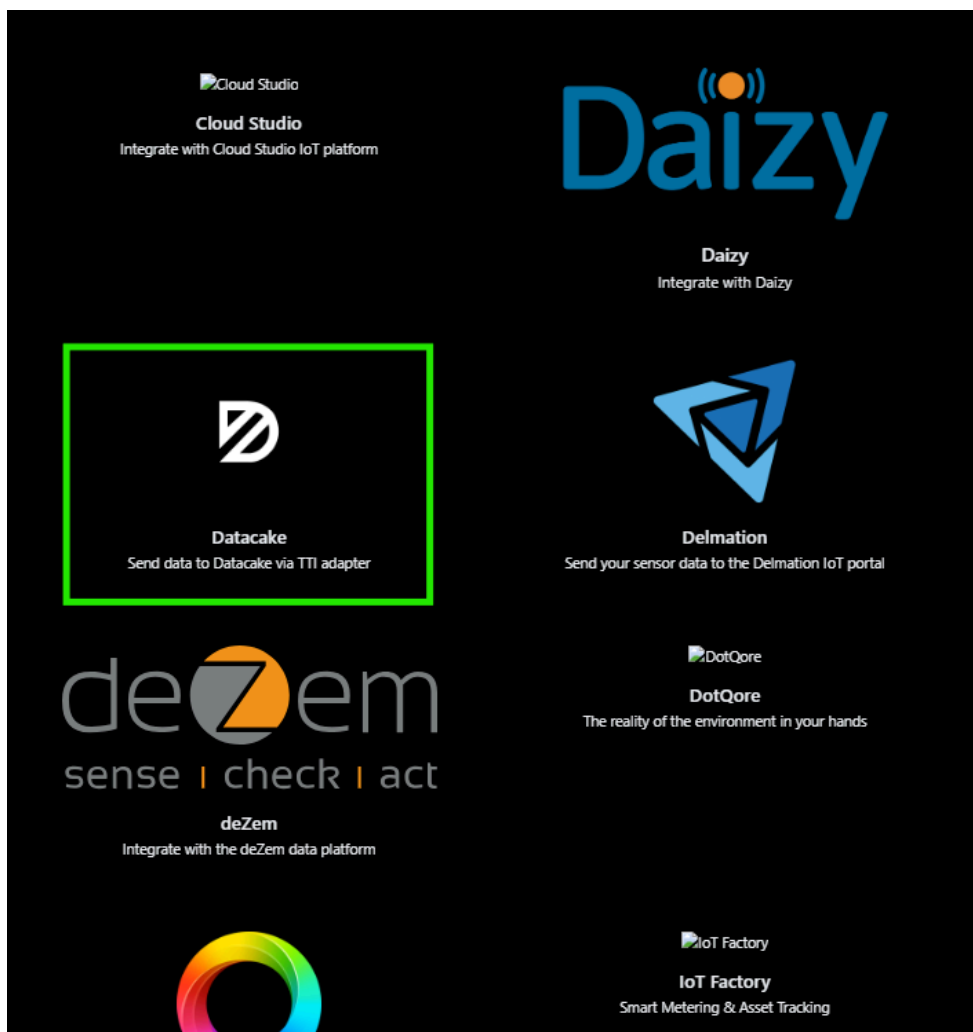
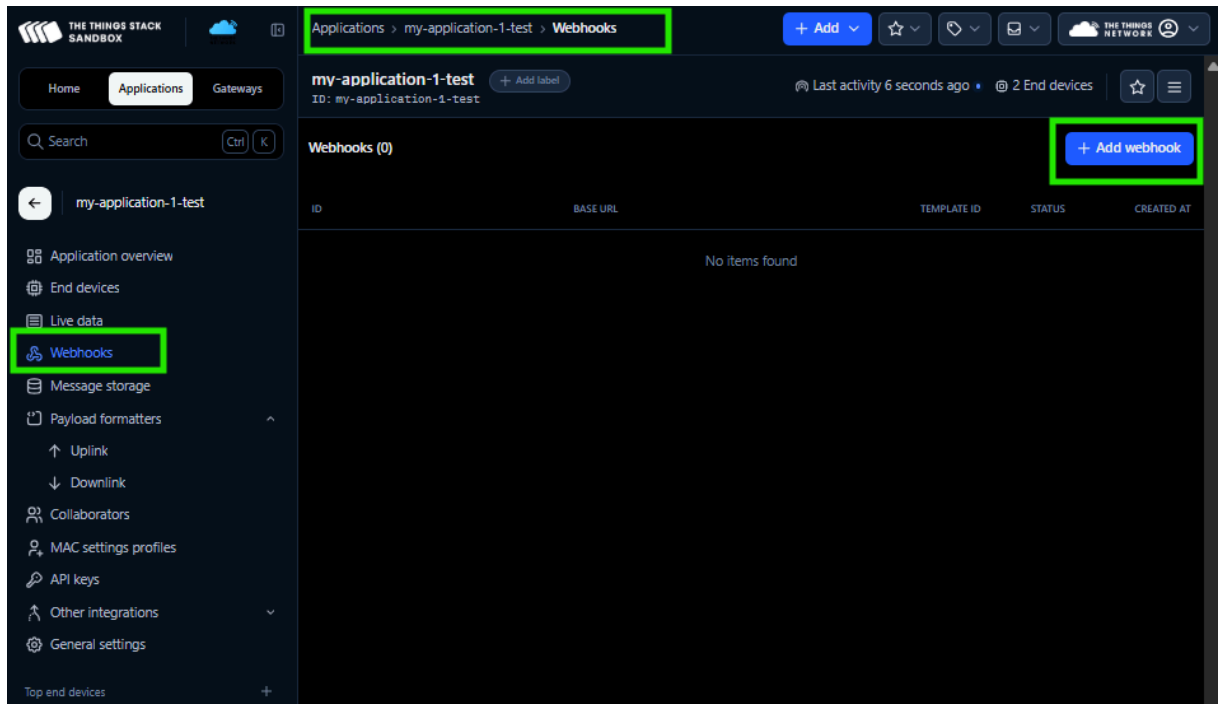


- Check if TTN is receiving the data; if so, turn off your development board.

TIME	TYPE	DATA PREVIEW	Verbose stream	Export as JSON	Settings
13:37:52	Forward uplink data message	DevAddr: 26 08 FC AF Payload: { battery_voltage: [-], humidity: [-], light: [-], pressure: [-], temperature: [-] } 00 73 27 10 01 67 01 04 02 68 64...	FFPort: 2 D		
13:37:52	Successfully processed data message	DevAddr: 26 08 FC AF			
13:37:41	Forward uplink data message	DevAddr: 26 08 FC AF Payload: { battery_voltage: [-], humidity: [-], light: [-], pressure: [-], temperature: [-] } 00 73 27 10 01 67 00 FA 02 68 64...	FFPort: 2 D		
13:37:41	Successfully processed data message	DevAddr: 26 08 FC AF			
13:37:33	Forward uplink data message	DevAddr: 26 08 FC AF Payload: { battery_voltage: [-], humidity: [-], light: [-], pressure: [-], temperature: [-] } 00 73 27 10 01 67 01 04 02 68 64...	FFPort: 2 D		
13:37:33	Successfully processed data message	DevAddr: 26 08 FC AF			
13:37:26	Forward uplink data message	DevAddr: 26 08 FC AF Payload: { battery_voltage: [-], humidity: [-], light: [-], pressure: [-], temperature: [-] } 00 73 27 10 01 67 01 04 02 68 64...	FFPort: 2 D		
13:37:26	Successfully processed data message	DevAddr: 26 08 FC AF			
13:37:16	Forward uplink data message	DevAddr: 26 08 FC AF Payload: { battery_voltage: [-], humidity: [-], light: [-], pressure: [-], temperature: [-] } 00 73 27 10 01 67 00 FA 02 68 64...	FFPort: 2 D		
13:37:16	Successfully processed data message	DevAddr: 26 08 FC AF			
13:37:09	Forward uplink data message	DevAddr: 26 08 FC AF Payload: { battery_voltage: [-], humidity: [-], light: [-], pressure: [-], temperature: [-] } 00 73 27 10 01 67 00 FA 02 68 64...	FFPort: 2 D		
13:37:09	Successfully processed data message	DevAddr: 26 08 FC AF			

Uplink forwarding TTN to Datacake

- Setting up webhooks



THE THINGS STACK
SANDBOX

Applications > my-application-1-test > Webhooks > Add > Datacake

my-application-1-test
ID: my-application-1-test

Home Applications Gateways

Search Ctrl K

my-application-1-test

- Application overview
- End devices
- Live data
- Webhooks
- Message storage
- Payload formatters
- Uplink
- Downlink
- Collaborators
- MAC settings profiles
- API keys
- Other integrations
- General settings

Setup webhook for Datacake
Send data to Datacake via TTI adapter
[About Datacake](#)

Webhook ID *

my-new-datacake-webhook-1

Token *

Datacake API Token

Create Datacake webhook

- You first need a Datacake account to get the token.
- Create your account in [Datacake](#).

- And add an "API User"

The screenshot shows the Datacake Administration interface. On the left is a sidebar with navigation options: Add Dashboard, Devices (All Devices, Manage Folders), Workspace (Gateways, Reports, Exports, Members, Rule Engine, Zones, Workspace, Integrations). The main content area is titled 'Members' and has tabs for 'In Workspace', 'Invited', and 'API Users'. The 'API Users' tab is selected. In the top right corner, there is a blue button labeled 'Add API User' which is highlighted with a green rectangle. Below the tabs, there is a table titled 'API Users' with columns: API USER, API KEY, and PERMISSIONS. The table contains two entries: 'usuario API Apiarior' and 'api-user-test-1'. Each entry has a 'Copy' and 'Show' button for the API key, and a list of permissions (Basics, Members, Devices, +7 for the first user; Devices for the second).

API USER	API KEY	PERMISSIONS
U usuario API Apiarior API	***** Copy Show	Basics Members Devices +7
A api-user-test-1 API	***** Copy Show	Devices

Add API User



Name *

api-user-test-1

Workspace Permissions

☒ Select All

☐ Basics

Can manage basics like name and logo

☐ Cake Red

Can manage and view Cake Red

☐ Rules

Can manage rules

☐ Reports

Can manage reports

☐ Zones

Can manage zones

☐ Members

Can manage members

☒ Devices

Can add / remove devices

☐ Gateways

Can manage gateways

☐ Dashboards

Can manage dashboards

☐ Exports

Can manage exports

Device Permissions

All devices in workspace



☒ Can view the device and its data

☐ Can edit device metadata

☐ Can edit the device definition

☒ Can record measurements

Add individual device permission

Search

Device name

Filter by tags

Match at least one tag

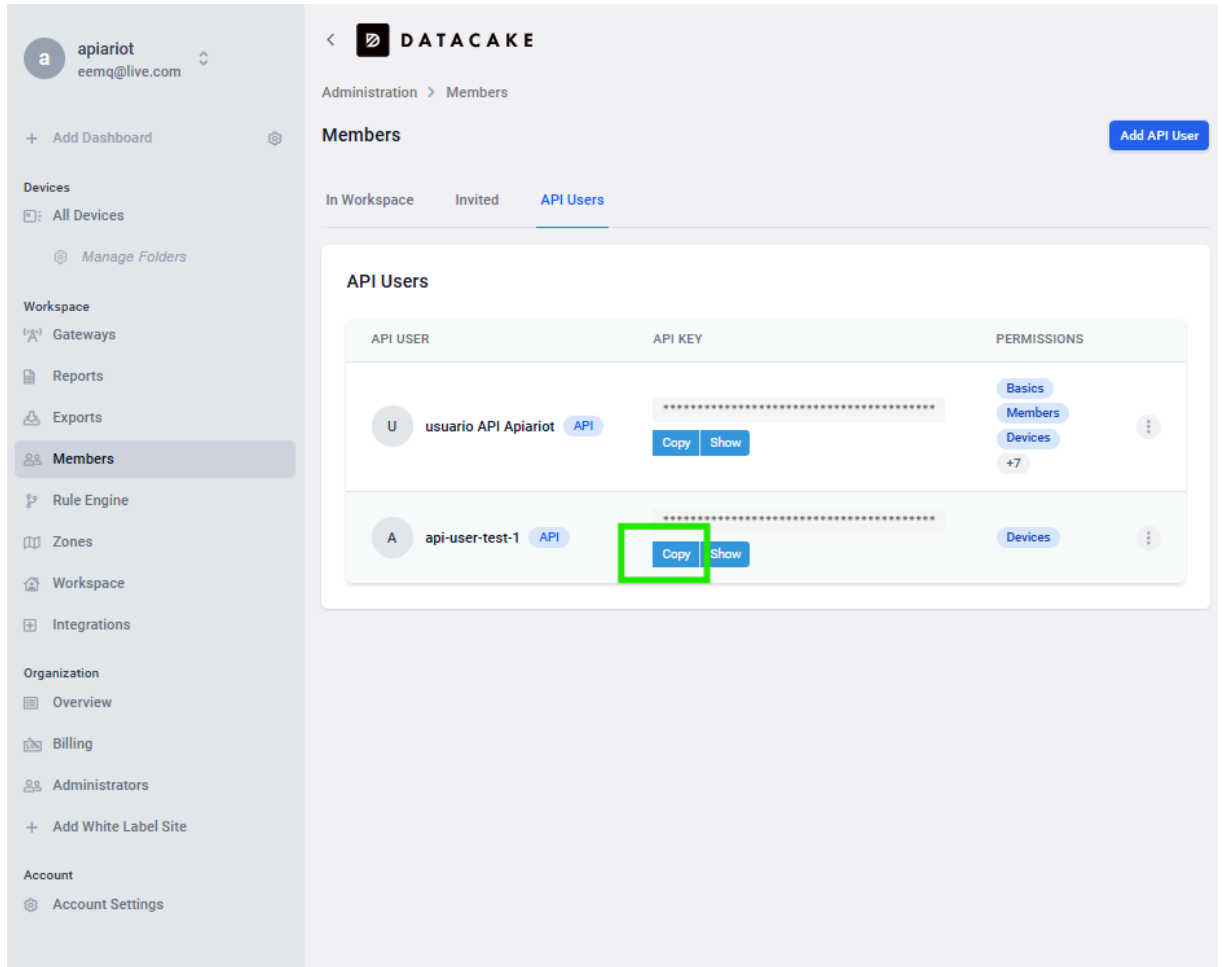


Search tags...

Cancel

Add API User

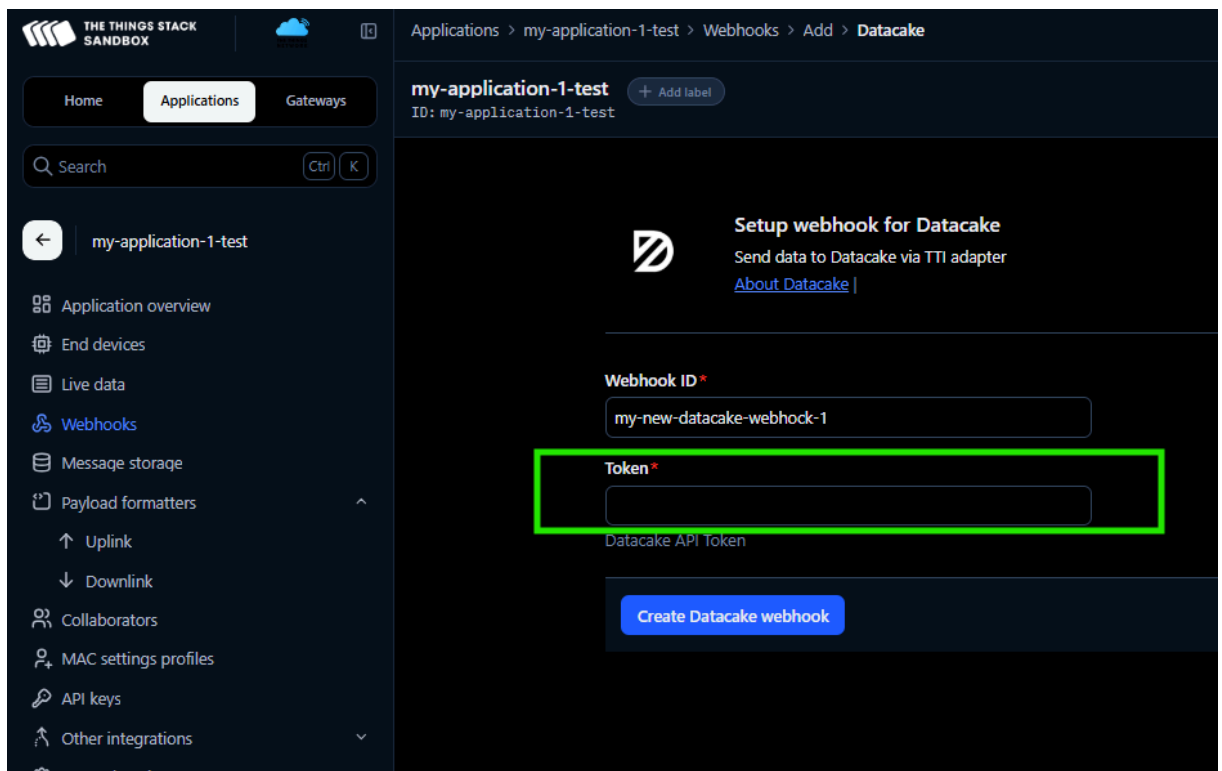
- Copy the token for TTN



The screenshot shows the Datacake interface. On the left is a sidebar with navigation options: Add Dashboard, Devices (All Devices, Manage Folders), Workspace (Gateways, Reports, Exports, Members, Rule Engine, Zones, Workspace, Integrations), Organization (Overview, Billing, Administrators, Add White Label Site), and Account (Account Settings). The main area is titled 'Members' and has tabs for 'In Workspace', 'Invited', and 'API Users'. The 'API Users' tab is selected, showing a table of API users. The table has columns for 'API USER', 'API KEY', and 'PERMISSIONS'. Two users are listed: 'usuario API Apiarlot' and 'api-user-test-1'. The 'Copy' button for the 'api-user-test-1' API key is highlighted with a green box.

API USER	API KEY	PERMISSIONS
U usuario API Apiarlot	***** Copy Show	Basics Members Devices +7
A api-user-test-1	***** Copy Show	Devices

- You now have the token for TTN



The screenshot shows the TTN (The Things Network) interface. The left sidebar has tabs for Home, Applications, and Gateways. The 'Applications' tab is selected, showing a list of applications. The 'my-application-1-test' application is selected, and the 'Webhooks' section is active. The 'Add' button is clicked, and the 'Datacake' webhook is being configured. The 'Webhook ID' is set to 'my-new-datacake-webhook-1'. The 'Token' field is highlighted with a green box, and the label 'Datacake API Token' is visible below it. The 'Create Datacake webhook' button is at the bottom.

Applications > my-application-1-test > Webhooks > Add > Datacake

my-application-1-test
ID: my-application-1-test

Setup webhook for Datacake
Send data to Datacake via TTN adapter
[About Datacake](#)

Webhook ID *
my-new-datacake-webhook-1

Token *
Datacake API Token

Create Datacake webhook

- Add your device now at Datacake

The screenshot displays the Datacake user interface. On the left is a sidebar with the user profile 'apiariot eemq@live.com' at the top. Below it are sections for 'Add Dashboard', 'Devices' (with 'All Devices' selected), 'Manage Folders', 'Workspace' (containing Gateways, Reports, Exports, Members, Rule Engine, Zones, Workspace, Integrations), 'Organization' (containing Overview, Billing, Administrators, and an 'Add White Label Site' link), and 'Account' (with 'Account Settings'). The main content area is titled 'Fleet > Devices'. It features a search bar, a 'Columns' dropdown, and a blue '+ Add Device' button highlighted with a green border. Below these are tabs for 'Overview', 'List' (selected), 'Grid', and 'Map'. A table header is visible with columns: 'DEVICE', 'PRIMARY', 'SECONDARY', 'DEVICE SIGNAL', and 'DEVICE BATTERY'. The table body contains a large warning icon and the text: 'There are no devices in your workspace, yet. Start by adding your first device using the button above.' At the bottom, it shows 'Showing 0 to 0 of 0 results', a '50 per page' dropdown, and a 'Previous' button.

Add Device

You have 4 devices left in your plan. ✕

First, choose the connectivity type of your device.

- ☒  **LoRaWAN**
Choose from 15 LoRaWAN networks
- ☐  **API**
Generic API device with support for MQTT and HTTP connectivity
- ☐  **Pincode Claiming**
Claim an existing device by pincode
- ☐  **Dragino NB-IoT**
Connect Dragino NB-IoT devices
- ☐  **1NCE**
Connect 1NCE devices

Next

- Use the network server: eu1.cloud.thethings.network (Switzerland)

Add LoRaWAN Device

You have 4 devices left in your plan. X

STEP 1
Product

STEP 2
Network Server

STEP 3
Devices

STEP 4
Plan

Datacake Product

You can add devices to an existing product on Datacake, create a new empty product or start with one of the templates. Products allow you to share the same configuration (fields, dashboard and more) between devices.

New Product from template

Create new product from a template

Existing Product

Add devices to an existing product

New Product

Create new empty product

Device Template

Datacake supports LoRaWAN devices from different manufacturers out of the box without complex configuration and setup.

caye

All Manufacturers

Cayenne LPP Starting Template

Datacake

Datacake

Back

Next

SCHRITT 1
Produkt

SCHRITT 2
Netzwerk-Server

SCHRITT 3
Geräte

SCHRITT 4
Plan

Netzwerk-Server

Bitte wählen Sie den LoRaWAN Network Server, mit dem Ihre Geräte verbunden sind.

☐		Datacake LNS AUTOMATIC SETUP Einfach starten und skalieren mit einem verwalteten LNS	Uplinks	Downlinks
☒		The Things Stack V3 TTN V3 / Things Industries	Uplinks	Downlinks
☐		netmore ThingPark	Uplinks	Downlinks
☐		KPN	Uplinks	Downlinks
☐		LORIoT	Uplinks	Downlinks
☐		ChirpStack	Uplinks	Downlinks
☐		Helium Verwenden Sie Ihre eigene Konsole	Uplinks	Downlinks
Anzeigen von 1 bis 6 von 14 Ergebnissen			Vorherige	Weiter

Zurück

Weiter

You have 5 devices left in your plan. ✕

Add LoRaWAN Device

STEP 1
Product

STEP 2
Network Server

STEP 3
Devices

STEP 4
Plan

Add Devices

Manual

Import from The Things Stack

Please provide one or multiple LoRaWAN device EUIs along with the corresponding names they should have on Datacake.

Alternatively, you can choose to upload a CSV file that contains the DevEUI, device Name, location, and a set of tags. For more information on how to format the file, please refer to [our documentation](#).

📎 Drag and drop a .csv file here or click to choose one

DEVEUI

NAME

LOCATION

TAGS



00 80 E1 15 05 00 D7 53

8 bytes

Name1

Location

Add tag

Ac

+ Add another device

Back

Next

You have 5 devices left in your plan. ✕

Add LoRaWAN Device

STEP 1
Product

STEP 2
Network Server

STEP 3
Devices

STEP 4
Plan

Your organization plan (5 / 5 remaining)

Free

7 days data retention
500 datapoints / day

Back

Add 1 device

- Turn your "End Node" back on.

The screenshot shows the Datacake web interface. On the left is a sidebar with navigation options: Add Dashboard, Manage Folders, Workspace (Gateways, Reports, Exports, Members, Rule Engine, Zones, Workspace, Integrations), and Organization. The main panel is titled 'Fleet > Devices' and shows a 'Devices' section with tabs for Overview, List, Grid, and Map. The 'List' tab is active, displaying a table with columns: DEVICE, PRIMARY, and SECONDARY. A single device is listed with ID '00.80.E1.15.05.00.D7.53', which is highlighted with a green box. The PRIMARY and SECONDARY columns show 'N/A'. Below the table, it says 'Showing 1 to 1 of 1 results'.

DEVICE	PRIMARY	SECONDARY
00.80.E1.15.05.00.D7.53	N/A	N/A

This screenshot shows the configuration page for the device '00.80.E1.15.05.00.D7.53' in the Datacake interface. The browser address bar shows 'app.datacake.de/apiariot/devices/d/a7b3b80b-1c1b-42d8-ac99-d23ef0f5ab4b/t/dashboard-0'. The page header indicates 'Fleet > 0080E115000AD87E'. The device ID '00.80.E1.15.05.00.D7.53' is displayed at the top. Below it, the 'Serial Number' is '0080E115000AD87E' and the 'Last update' is 'Sun Sep 27, 2026 21:38:05 GMT+02:00'. A navigation bar at the bottom of the main panel includes 'Dashboard', 'History', 'Downlinks', 'Configuration' (highlighted with a green box), 'LoRaWAN', 'Debug', and 'Permissions'.

- Wait until data is received.

apiariot
eemq@live.com

+ Add Dashboard

Devices

All Devices

Manage Folders

Workspace

Gateways

Reports

Exports

Members

Rule Engine

Zones

Workspace

Integrations

Organization

Overview

Billing

Administrators

+ Add White Label Site

Account

Account Settings

Support

Documentation

Changelog

Basic device settings and metadata

Product & Hardware

Product settings and hardware-specific configuration

Fields

Configure measurement fields and data structure

Integrations

Third-party connections and data exports

Danger Zone

Device deletion and destructive actions

+ Add Field

Fields

Fields describe the data the device will store.

Live data

NAME	IDENTIFIER	TYPE	ROLE / SEMANTIC	CURRENT VALUE	LAST UPDATE
Temperature Sensor 0	TEMPERATURE_SENSOR_0	Numeric		0 °C	1 minute ago
Analog Input 0	ANALOG_INPUT_0	Numeric		0	1 minute ago
GPS 1	GPS_1	Location	Location	0.000000, 0.000000	1 minute ago

Suggested Fields

The device's decoder returned fields that have not yet been created on the product. You can review and create them from here.

Delete All Suggestions

IDENTIFIER	GUESSED TYPE	FIRST SEEN	LAST SEEN	LAST VALUE	
TEMPERATURE_SENSOR_1	Numeric	4 minutes ago	2 minutes ago	22	Create Field
HUMIDITY_2	Numeric	4 minutes ago	2 minutes ago	50	Create Field
BAROMETER_0	Numeric	4 minutes ago	2 minutes ago	1.000	Create Field

Configuration Fields

Configuration fields hold a static value and can have a product-wide default value, that can be overwritten on a device level. They can be accessed in decoders.

+ Add Configuration Field

NAME	IDENTIFIER	FIELD TYPE	DESCRIPTION	VALUE
------	------------	------------	-------------	-------

No fields have been created, yet

Create configuration fields to define configuration variables.

×

Add Field

Fields define the schema of the data the device stores.

Type

Numeric

Name

Temperature Sensor 1

Identifier

TEMPERATURE_SENSOR_1

The field identifier is a unique string that can consist of uppercase letters, numbers and underscores. Once a field has been created, the identifier can not be changed.

Unit

Optional

Role

None

Assign roles to highlight key measurement fields across the platform; each role is limited to one field per product.

Semantic

None

Assign semantic fields to standardize data across devices.

Formula

Optional

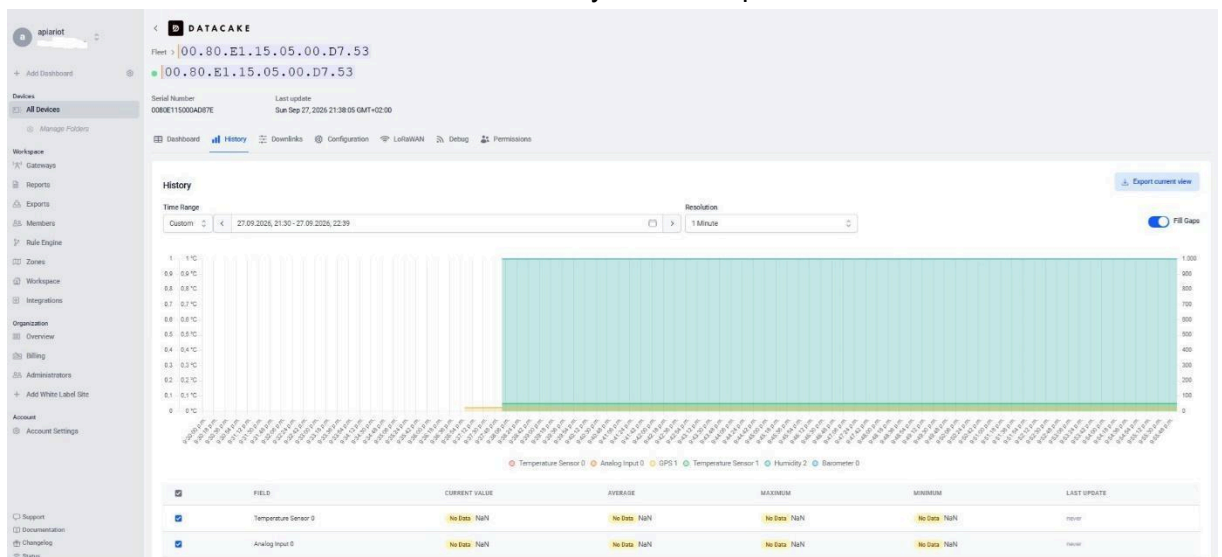
Formulas can be used to perform calculations on values from other fields. Fields that have a formula can not be written to from a decoder or via the API.

☐ Use Formula

Cancel

Add Field

- You should now be able to see the data from your development board in Datacake.



- And in TTN as well

The screenshot shows the TTN console interface for a device named 'device-3-d87e'. The 'Live data' tab is selected, displaying a list of events. The events are as follows:

TIME	TYPE	DATA PREVIEW
15:39:06	Schedule data downlink for transmission	DevAddr: 26 08 C7 5C Rx Delay: 5
15:39:06	Forward uplink data message	DevAddr: 26 08 C7 5C Payload: { battery_voltage: [-], humidity: [-], light: [-], pressure: [-], temperature: [-] }
15:39:06	Successfully processed data message	DevAddr: 26 08 C7 5C
15:39:06	Schedule data downlink for transmission	DevAddr: 26 08 C7 5C Rx Delay: 5
15:39:06	Forward uplink data message	DevAddr: 26 08 C7 5C Payload: { battery_voltage: [-], humidity: [-], light: [-], pressure: [-], temperature: [-] }
15:39:06	Successfully processed data message	DevAddr: 26 08 C7 5C
15:39:40	Forward join-accept message	DevAddr: 26 08 C7 5C JoinEUI: 00 00 E1 15 05 30 FC 63 DevEUI: 00 00 E1 15 05 30 FC 63
15:39:39	Successfully processed join-request	DevAddr: 26 08 AD 2C JoinEUI: 00 00 E1 15 05 30 FC 63 DevEUI: 00 00 E1 15 05 30 FC 63
15:39:38	Accept join-request	DevAddr: 26 08 C7 5C JoinEUI: 00 00 E1 15 05 30 FC 63 DevEUI: 00 00 E1 15 05 30 FC 63

At the bottom of the event list, a message states: 'Console: Events cleared. The events list has been cleared.'